Information Fusion 127 (2026) 103919



Contents lists available at ScienceDirect

Information Fusion

journal homepage: www.elsevier.com/locate/inffus



Full Length Article



Generative recommender systems: A comprehensive survey on model, framework, and application

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ARTICLE INFO

Keywords:
Recommender systems
Generative recommendation
Generative models
Feature representation
Large language models (LLMs)
Multimodal
Information fusion

ABSTRACT

With the exponential growth of information in the Internet era, Recommender Systems play a pivotal role in helping users discover content of interest. However, traditional recommendation paradigms rely heavily on large-scale human-annotated data and still face limitations in expressive power and content generation when dealing with semantically rich and context-sensitive recommendation tasks due to the intricate feature engineering. To overcome these challenges, generative models, owing to their powerful semantic modeling and content generation capabilities, have been integrated into recommender systems, leading to the emergence of Generative Recommender Systems (GRS). GRS can better understand user preferences and generate diverse and highly personalized recommendation content, significantly enhancing user experience and recommendation effectiveness. This paper presents a comprehensive and in-depth survey of GRS. Distinct from previous surveys, our work provides comprehensive coverage of both Large Language Models (LLMs)-driven recommendation paradigms and applications of other mainstream generative models in recommendation tasks. In addition, we introduce a novel perspective to systematically categorize the key technical paths of GRS, exploring current research advances across multiple dimensions, including data representation, model architecture, information fusion, and evaluation methodologies. Furthermore, this paper summarizes the critical engineering challenges and practical approaches encountered during the industrial deployment of GRS, discussing their technical advantages and limitations in typical industrial scenarios. Finally, we provide insight into future research directions in this field, offering a more comprehensive perspective and a deeper understanding to relevant researchers and practitioners.

1. Introduction

Recommender systems serve as a critical bridge between users and information, playing a central role in modern information service platforms [1]. Traditional recommender systems primarily rely on discriminative modeling approaches such as collaborative filtering (CF) [2] and content-based recommendation (CBR) [3]. These methods aim to predict user preferences for unseen items based on existing user-item interactions and subsequently generate ranked recommendations, but face significant limitations in their expressive power and content generation capabilities. With the advancement of deep learning, embedding-based approaches [4,5] have further enhanced the expressive power of recommender models, enabling more accurate capture of latent user interests and semantic item features.

Despite these improvements, traditional methods often face limitations, including data sparsity, cold-start problems, and insufficient contextual modeling, which hinder their ability to meet the growing demand for personalized and diverse recommendations. In response, generative models have recently gained attention due to their breakthroughs in natural language processing [6], image generation [7], and

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https://doi.org/10.1016/j.inffus.2025.103919
Received 2 July 2025; Received in revised form 9 October 2025; Accepted 5 November 2025
Available online 9 November 2025
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Table 1
Comparison between different surveys. "AE" denotes AutoEncoder, "AR" denotes AutoRegressive model.

Year	Papers	AE	AR	GAN	Diffusion	LLM	Representation	Information Fusion	Industrial
2020	[20]	✓							
2024	[21]	✓							
2021	[22]			✓					
2024	[24]				✓				
2025	[23]				✓				
2023	[14]					✓			
2024	[13]					✓	✓		
2024	[16]					✓			
2024	[17]					✓			
2024	[25]					✓	✓		✓
2024	[15]					✓	✓		✓
2024	[19]					✓	✓		
2024	[18]	✓	✓	✓	✓	✓			
2025	**Ours**	✓	✓	✓	✓	✓	✓	✓	✓

other domains [8,9]. To overcome fundamental challenges and move beyond simple item ranking to truly personalized content generation, generative models have been integrated into recommender systems, bringing a new paradigm of Generative Recommender Systems (GRS). GRS can better understand user preferences and directly generate diverse and highly personalized recommendation content, significantly enhancing user experience and recommendation effectiveness. This approach overcomes the constraints of traditional discriminative models that rely solely on observed interaction data. Moreover, the emergence of Large Language Models (LLMs) has further expanded the capabilities of GRS, endowing them with stronger semantic understanding and cross-modal reasoning abilities. These models can effectively integrate multiple information sources, such as text and images, to improve personality and user satisfaction.

Recently, research in GRS has explored various model architectures, including Variational AutoEncoder (VAE) [10], AutoRegressive models [6,8,11], Generative Adversarial Networks (GANs) [9], and diffusion models [12]. These approaches demonstrate distinct advantages across different recommendation tasks, such as mitigating data sparsity, enabling end-to-end recommendation generation, and enhancing recall performance for long-tail items. However, several challenges remain, including the stability of recommendation generation in cold-start scenarios, effective multimodal information fusion, and computational efficiency bottlenecks during training and inference. These issues require further exploration from both theoretical and engineering perspectives.

While several studies [13–19] have attempted to analyze and summarize GRS, they often focus on individual model architectures or specific application scenarios, lacking a comprehensive comparative analysis across different generative approaches. Most existing studies examine methodologies within isolated frameworks, failing to comprehensively understand the feature processing and information fusion mechanisms of GRS, as well as the application challenges and solution strategies in industrial scenarios. These limitations hinder the development of more effective GRS solutions and constrain the field's ability to establish clear research directions and unified best practices.

As shown in Table 1, current survey efforts remain fragmented, while some studies [13–17] systematically analyze LLM-empowered recommendation methods, they narrowly focus on LLM-specific techniques; others [18] compare generative versus traditional recommendation paradigms but fail to trace the technical evolution across broader generative approaches. Though Huang et al. [19] provide domain-specific reviews on foundation model adaptation tasks, they lack a unified theoretical framework for analyzing task innovation mechanisms. Moreover, some works have individually reviewed the applications of recommender systems in specific categories of generative models, such as AutoEncoder (AE) models [20], Variational AutoEncoder (VAE) models [21], Generative Adversarial Network (GAN) models [22], and Diffusion models [23,24]. However, these works primarily focus on analyzing methodologies within individual model frameworks. Despite the growing interest in leveraging generative models for recommendation tasks, there remains a lack of comprehensive analysis that covers the overall workflow of generative recommender systems, as well as the diverse applications and characteristics of different generative model types within this domain.

This paper presents a comprehensive and in-depth survey of GRS, which is designed to fill a critical gap by providing a unique perspective that is both more comprehensive and systematic than prior reviews. This structured approach provides a clear roadmap of the field's evolution, highlights the core differences from traditional recommendation systems, and offers valuable insights into future research directions. The contributions of this survey are summarized as follows:

1. We establish a novel workflow perspective for GRS, systematically categorizing mainstream generative architectures as complementary paradigms. Our analysis delineates their technical characteristics, applicable scenarios, and optimization directions, offering structured insights into their capabilities and limitations.
2. We comprehensively examine foundational data types and trace the technical evolution of feature representation methods. Furthermore, we investigate advanced cross-modal and interaction-aware fusion techniques, proposing tailored integration strategies for handling multi-source heterogeneous data in GRS.
3. We construct a holistic evaluation framework encompassing conventional metrics and generative-specific dimensions. Additionally, we identify critical engineering challenges, bridging the gap between academic and industrial deployment.
4. Finally, we identify and discuss the current technical bottlenecks faced in the GRS field, highlight potential research avenues for each challenge, and outline promising future directions for advancing GRS.

The remaining sections are organized as follows: Section 2 introduces data sources and utilization patterns in generative recommender systems. Section 2 examines paradigms of recommendation models based on different generative architectures and their adaptability across recommendation tasks. Section 4 explores information fusion mechanisms for integrating multi-source heterogeneous data in cross-modal semantic and collaborative modeling. Section 5 discusses evaluation frameworks, covering both traditional metrics and newly proposed assessment methods for generative tasks. Section 6 analyzes application challenges and technical issues through representative industrial scenarios. Section 7 explores future research directions and emerging trends in the field. Finally, Section 8 concludes by summarizing major contributions and identifying promising avenues for further investigation. The overall structure of the paper is shown in Fig. 1.

2. Data sources and feature representation

The core objective of generative recommender systems is to model the latent relationships between user behaviors and item characteristics, enabling the generation of personalized recommendation results. Therefore, the performance of such systems heavily relies on the quality, diversity, and representational effectiveness of the input data. Understanding the data sources and feature extraction methods used in generative recommender systems is essential for building efficient and interpretable models. This section begins with an overview of commonly used public datasets in generative recommender systems, followed by a detailed discussion on the primary sources and forms of data, including various data types. Subsequently, we analyze the evolution of feature representation techniques from traditional ID-based representations to semantic description, product quantization, residual quantization, and LLM tokenization, highlighting each method's technical development and application scenarios. This foundation will support the modeling paradigms discussed in later sections.

2.1. Overview of commonly used datasets

The research and application of recommender systems heavily rely on the support of high-quality datasets. Different types of datasets reflect the diversity of user behavior patterns and determine the modeling approaches and task designs. Currently, commonly used datasets can be broadly categorized into five types: general recommendation datasets, sequential recommendation datasets, multimodal recommendation datasets, conversational recommendation datasets, and multi-behavior recommendation datasets, each with distinct data structures and application scenarios. Commonly used datasets and their category information are shown in Table 2.

General recommendation datasets primarily consist of user ratings or click records, emphasizing static interactions between users and items. These datasets are widely used for training and evaluating basic collaborative filtering algorithms.

- **Yelp Challenge** [26]: The Yelp dataset includes 4.7M reviews, 150K businesses, 200K images, and data from 12 metropolitan areas, covering business attributes and user check-ins.
- **ZhihuRec** [27]: A large-scale real-world interaction dataset collected over 10 days from Zhihu, a leading Chinese knowledge-sharing platform. It comprises approximately 100 million user interactions, involving 798,000 users, 165,000 questions, 554,000 answers, 240,000 authors, and 70,000 topics. Additionally, it includes over 501,000 user search queries. The dataset also provides anonymized textual descriptions of users, answers, questions, authors, and topics. It is currently the largest publicly available dataset for personalized recommendation research based on real-world interactions.
- **MIND** [28]: A news recommendation dataset comprises 160,000 news articles and 24 million user interactions, including 15 million impression records, with detailed textual features spanning article titles, abstracts, and full body content.

Sequential recommendation datasets focus on the temporal dynamics of user behavior by recording sequential actions, including clicks, views, or purchases over time. These datasets support modeling the evolution of user interests and are commonly used to evaluate the capabilities of autoregressive generative models.

- **Tenrec** [29]: A multi-scenario dataset with 5M users and 140M interactions across articles/videos, including true negative feedback and demographic features.
- **MovieLens** [30]: A movie rating dataset containing up to 25 million ratings on 27,000 movies, user demographic information, and movie genre metadata.
- **Douban** [31]: A movie dataset formatted in the MovieLens style, contains 2.8 million ratings provided by 28,000 users on a collection of 5000 movies, making it suitable for cross-platform recommendation studies.
- **Amazon Review** [32]: A product reviews dataset contains 233.1 million user comments across 18 product categories, including both

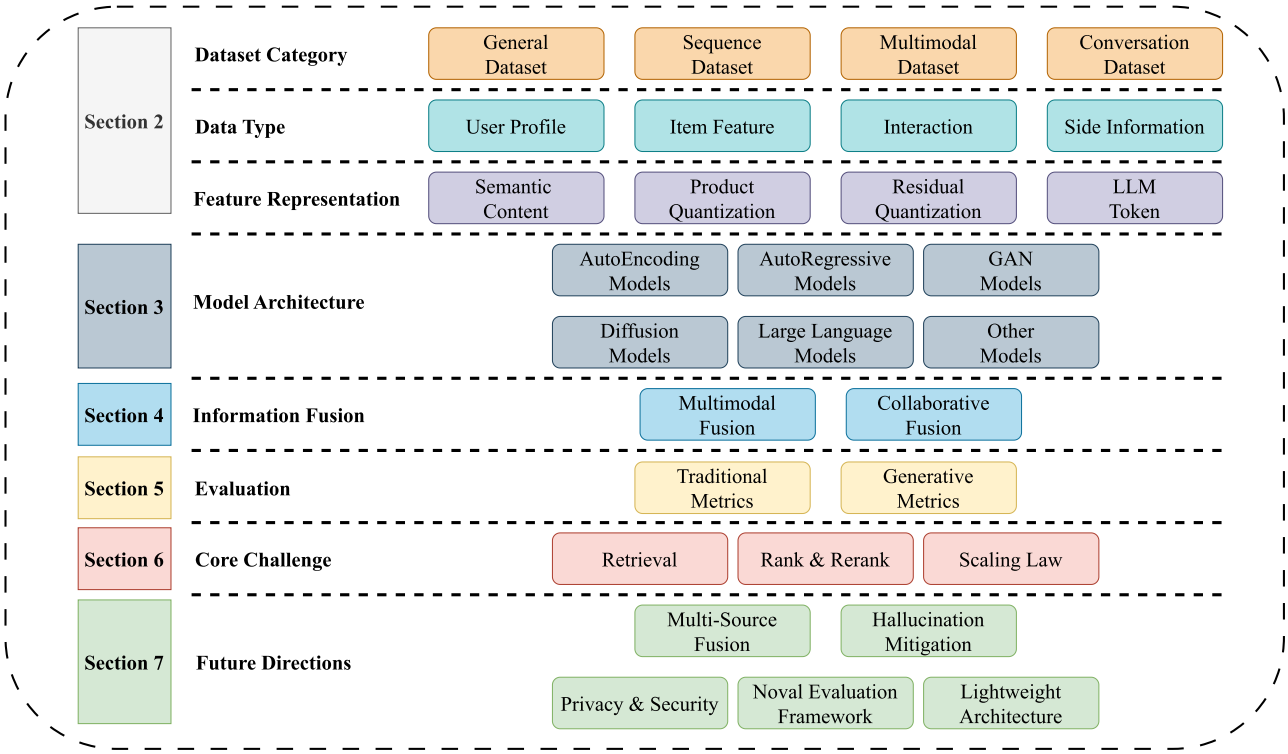


Fig. 1. Overall structure of the paper.

Table 2
Typical benchmark datasets for generative recommender systems.

Scene	Tasks	Dataset	Release Year*
General	Commerce	Yelp Challenge [26][1]	2018
	Q&A	ZhihuRec [27][3]	2021
	News	MIND [28][2]	2020
Sequence	Multi Scenes	Tenrec [29][4]	2022
	Movies	Movielens [30][5]	1998–2024
		Douban [31][6]	2017
	Products	Amazon Review [32][7]	2013–2018
		Online Retail [33][8]	2012
	Entertainment	LastFM [34][9]	2011
		Steam [35][10]	2022
Multimodal	Short Videos	Youtube [36][11]	2018–2019
		KuaiRec [37][12]	2022
		KuaiSAR [38][13]	2023
		MicroLens [39][14]	2023–2025
	Images	PixelRec [40][15]	2024
Conversation	Movies	ReDial [41][16]	2018
		INSPIRED [42][17]	2020
	Products	AmazonM2 [43][18]	2023
		U-need [44][19]	2023
Multi-Behavior	E-commerce	IJCAI[20]	2015
		Taobao[21]	2018
		Tmall[22]	2022

Data availability: [1] https://www.yelp.com/dataset,
[2] https://msnews.github.io/,
[3] https://github.com/THUIR/ZhihuRec-Dataset,
[4] https://tenrec0.github.io/,
[5] https://grouplens.org/datasets/movielens/,
[6] https://github.com/MarkWuNLP/MultiTurnResponseSelection,
[7] https://nijianmo.github.io/amazon/index.html,
[8] https://archive.ics.uci.edu/dataset/352/online+retail,
[9] http://millionsongdataset.com/lastfm/#getting,
[10] https://www.kaggle.com/datasets/fronkongames/steam-games-dataset,
[11] https://research.google.com/youtube8m/,
[12] https://kuairec.com/,
[13] https://kuaisar.github.io/,
[14] https://github.com/westlake-repl/MicroLens/,
[15] https://github.com/westlake-repl/PixelRec,
[16] https://redialdata.github.io/website/,
[17] https://github.com/sweetpeach/Inspired,
[18] https://kddcup23.github.io/#dates,
[19] https://github.com/LeeeeoLiu/U-NEED,
[20] https://ijcai-15.org/repeat-buyers-prediction-competition/,
[21] https://tianchi.aliyun.com/dataset/649,
[22] https://tianchi.aliyun.com/dataset/140281.
Note: * Year spans reflect dataset version updates; single years indicate initial releases.

numerical ratings on a scale from 1 to 5 and textual reviews, making it suitable for sequence-based recommendation tasks.

- **Online Retail** [33]: A transactional dataset that records all transactions occurring between December 1, 2010, and December 9, 2011, for a UK-based and registered non-store online retail company. The dataset includes detailed information such as customer ID, product description, quantity, unit price, invoice date, and stock code, amounting to 541,909 instances.
- **LastFM** [34]: A music dataset released in 2011 in conjunction with the Million Song Dataset (MSD), featuring the most extensive collection of song-level tags. It primarily includes audio features of the songs in the MSD catalog, along with artist information and lyrics. The dataset comprises 584,897 tracks, 522,366 unique tags, 8,598,630 track-tag pairs, and 56,506,688 similar track-track pairs.
- **Steam** [35]: A game dataset of 97,410 Steam games with reviews, prices, and tags for popularity analysis and sales prediction.

Multimodal recommendation datasets go beyond traditional collaborative signals by integrating unstructured modalities like text reviews, images, and audio. These datasets aim to enhance semantic understanding and cross-modal reasoning capabilities, providing foundational support for LLM-driven and multimodal fusion-based generative recommender systems.

- **Youtube** [36]: The YouTube-8M dataset introduced by Google Research in 2016 is a large-scale labeled video dataset comprising over 6.1 million YouTube video IDs. Each video is annotated with high-quality, machine-generated labels from a diverse vocabulary of 3862 visual entities. The dataset contains over 350,000 hours of video content and approximately 2.6 billion extracted visual and audio features. The average duration of each video segment is around 5 seconds.
- **KuaiRec** [37]: A real-world dataset collected from the recommendation logs of the video-sharing mobile app Kuaishou. It includes interaction behaviors of 1411 users with 3327 short videos from July 5, 2020, to September 5, 2020.
- **KuaiSAR** [38]: A short-video dataset featuring pre-extracted behavioral and implicit feedback signals, without inclusion of raw content, and with unspecified overall size.
- **MicroLens** [39]: A large-scale short-video dataset containing 1 billion user interactions, 34 million users, 1 million videos, and raw multimodal features for each item.
- **PixelRec** [40]: A video cover recommendation dataset containing 200 million user interactions, 30 million users, and 400,000 images.

Conversational recommendation datasets typically include dialogue histories between users and recommendation agents. These datasets emphasize context-aware interest modeling and intent recognition, making them suitable for developing interactive generative recommender systems and promoting the integration of conversational recommendation with natural language generation.

- **ReDial** [41]: An annotated dialogue dataset where users recommend movies to each other. The dataset contains over 10,000 dialogues centered around movie recommendations.
- **INSPIRED** [42]: A dataset of 1001 human-to-human conversations in the movie recommendation domain.
- **Amazon M2** [43]: The dataset constitutes a large-scale collection of e-commerce user sessions and item interactions, encompassing 1,410,675 products distributed across six distinct regions.
- **U-NEED** [44]: A user-centered conversational recommendation dataset for e-commerce, containing 7698 fine-grained annotated pre-sales conversations, 333,879 user behaviors, and 332,148 item knowledge tuples.

Multi-behavior recommendation datasets contain various types of user-item interactions beyond explicit ratings, such as views, clicks, add-to-cart, and purchases. These datasets are primarily collected from e-commerce platforms, where users generate rich behavioral traces during their shopping journeys. They capture diverse and sequential user activities, enabling more accurate modeling of user preferences and behavior patterns.

- **IJCAI**: An e-commerce recommendation dataset released for the IJCAI 2015 Challenge. It contains user behaviors such as clicks, purchases, add-to-cart, and favorites.
- **Taobao**: A large-scale e-commerce dataset collected from Alibaba's Taobao platform, consisting of behavioral records from approximately one million randomly sampled users. It includes various user actions such as clicks, purchases, add-to-cart, and favorites. Each record contains user ID, item ID, category ID, behavior type, and timestamp.
- **Tmall**: A multi-behavior e-commerce dataset collected from Alibaba's Tmall platform, consisting of user interactions including clicks, collections, add-to-cart, and purchases. It includes over 8.1 million items and captures rich behavioral traces from hundreds of thousands of users.

2.2. Data sources and main forms

In generative recommender systems, high-quality data is the foundation for modeling user preferences, item characteristics, and generating personalized recommendation results. Based on their semantic attributes and modeling purposes, the data sources of recommender systems can be categorized into three major types: user profiles, item features, and user-item interaction behaviors. These data can also be classified into static and dynamic data, reflecting the evolution of long-term properties and short-term interest of users and items.

2.2.1. User profiles

User profiles encompass descriptive user information, including basic demographic attributes, long-term interest tags, and behavioral patterns. Basic attributes typically consist of structured data such as age, gender, occupation, and registration details, which fall into the category of static data. In contrast, interest tags derived from historical behavior, such as recently browsed product categories or preferred brands, are dynamic and reflect evolving user preferences.

2.2.2. Item features

Item features are used to characterize the attributes of recommended objects and serve as a crucial basis for recommender systems to understand item properties and uncover user interests. These features are typically categorized into two types: structured and unstructured. Structured features, such as category labels, price, brand, and inventory status, provide clear semantic descriptions and statistical information. Unstructured features, on the other hand, include multimodal content such as titles, descriptive texts, cover images, audio, and video, which enhance the model's ability to understand the semantic information of items.

2.2.3. User-item interaction behaviors

User-item interactions constitute the core signals in recommender systems, reflecting users' preference patterns and behavioral tendencies. Common interaction data include clicks, views, favorites, add-to-cart actions, purchases, ratings, comments, and shares. These behaviors can be categorized into explicit feedback, such as ratings, favorites, likes, and implicit feedback, such as dwell time, repeated plays, and scrolling depth. Explicit feedback typically carries clear semantic meaning and directly indicates user sentiment toward an item. In contrast, although lacking explicit emotional labels, implicit feedback is more abundant in quantity and provides a more comprehensive view of user behavior trajectories. In recent years, increasing research efforts have focused on joint modeling of multiple behavior types to more precisely capture the dynamic evolution of user interests, particularly in sequential and generative recommendation settings.

From a temporal perspective, basic user attributes and item metadata generally belong to static data, characterized by their relatively stable nature. On the other hand, recent user behaviors, interaction sequences, and conversation logs fall under dynamic data, which better reflect real-time changes in user interests. Consequently, dynamic data plays a more critical role in generative recommender systems, especially in architectures that heavily rely on temporal information, such as autoregressive generation and diffusion modeling. These diverse data types constitute the foundational input sources for generative recommender systems, providing rich semantic and behavioral bases for subsequent feature representation learning and generative modeling.

2.3. Evolution of feature representation methods

The development of generative recommender systems has witnessed significant evolution in feature representation methodologies, progressing from early ID-based discrete encoding to vector-quantized semantic modeling, and ultimately to contemporary LLM-driven unified representation generation. This technological progression has substantially enhanced the expressive power of recommender systems. Such advancement addresses two fundamental requirements: (1) overcoming the inherent limitations of traditional collaborative filtering in handling data sparsity and cold-start scenarios, and (2) establishing more generalizable cross-domain representation frameworks to facilitate representation space alignment for generative recommendation models. This section systematically reviews the evolutionary trajectory, particularly emphasizing how technological breakthroughs at each stage have progressively enriched semantic comprehension capabilities in generative recommender systems. The classification of feature representation methods is shown in Fig. 2. For a comprehensive analysis of different feature representation approaches, Table 3 presents the Normalized Discounted Cumulative Gain (NDCG) performance of selected methods in various category-specific subsets of the Amazon Review dataset.

2.3.1. Semantic content representation

Traditional ID-based recommender systems typically map raw user and item IDs into embedding vectors, which are then processed by deep learning models such as the deep learning recommendation model (DLRM) to learn feature representations. However, this approach mainly relies on categorical features associated with user and item IDs, lacking intrinsic semantic information. As a result, it struggles to capture associations between similar users or items, limiting the model's generalization capability in cold-start scenarios. Moreover, in large-scale recommender systems, the assignment of unique identifiers leads to an explosion in the number of IDs. Furthermore, continuously introducing new items can cause ID drift and bias, further degrading model performance over time.

Generative recommender systems have explored constructing semantically aware discrete tokens from textual and attribute features of items to better capture the rich semantic information inherent in recommendation content. Hua et al. [45] proposed a content-based semantic indexing method that builds target IDs from item metadata by organizing categories into a multi-level hierarchical structure. Each non-leaf node represents a category in this structure, while each leaf node corresponds to an individual item. The index for each item is formed by concatenating the path from root to leaf, thereby incorporating both coarse-grained and fine-grained category information. Hou et al. [46] introduced the first context-aware tokenization approach that dynamically identifies co-occurrence patterns across feature sequences within a single set and neighboring sets, merging them into new tokens to construct a semantically enriched vocabulary.

A key challenge in generative recommendation lies in mapping items into a linguistic feature space that aligns with language models. Lin et al. [47] proposed a multi-level identifier that integrates numeric IDs, item titles, and attributes to represent items, balancing discriminability and semantic expressiveness jointly. They also designed a specialized data structure to ensure only valid identifiers are generated, enabling the ranking of items within the corpus using these identifiers. Qiu et al. [48] introduced ControlRec, a contrastive prompt learning framework that aligns textual semantics with item IDs and constructs a shared representation space to bridge the semantic gap between LLMs and item representations. Jin et al. [49] employed a text tokenizer to tokenize each item's textual description and concatenated the resulting tokens into a sequence input, allowing the model to learn semantic ID representations self-supervisedly. Zhai et al. [50] treated user interaction behaviors as sequential actions of the model, interleaving <item, action> token pairs at each interaction step for training purposes. These efforts aim to overcome the limitations of traditional ID-based systems by enriching item representations with semantic meaning, thereby enhancing generative recommender systems' expressiveness and generalization ability.

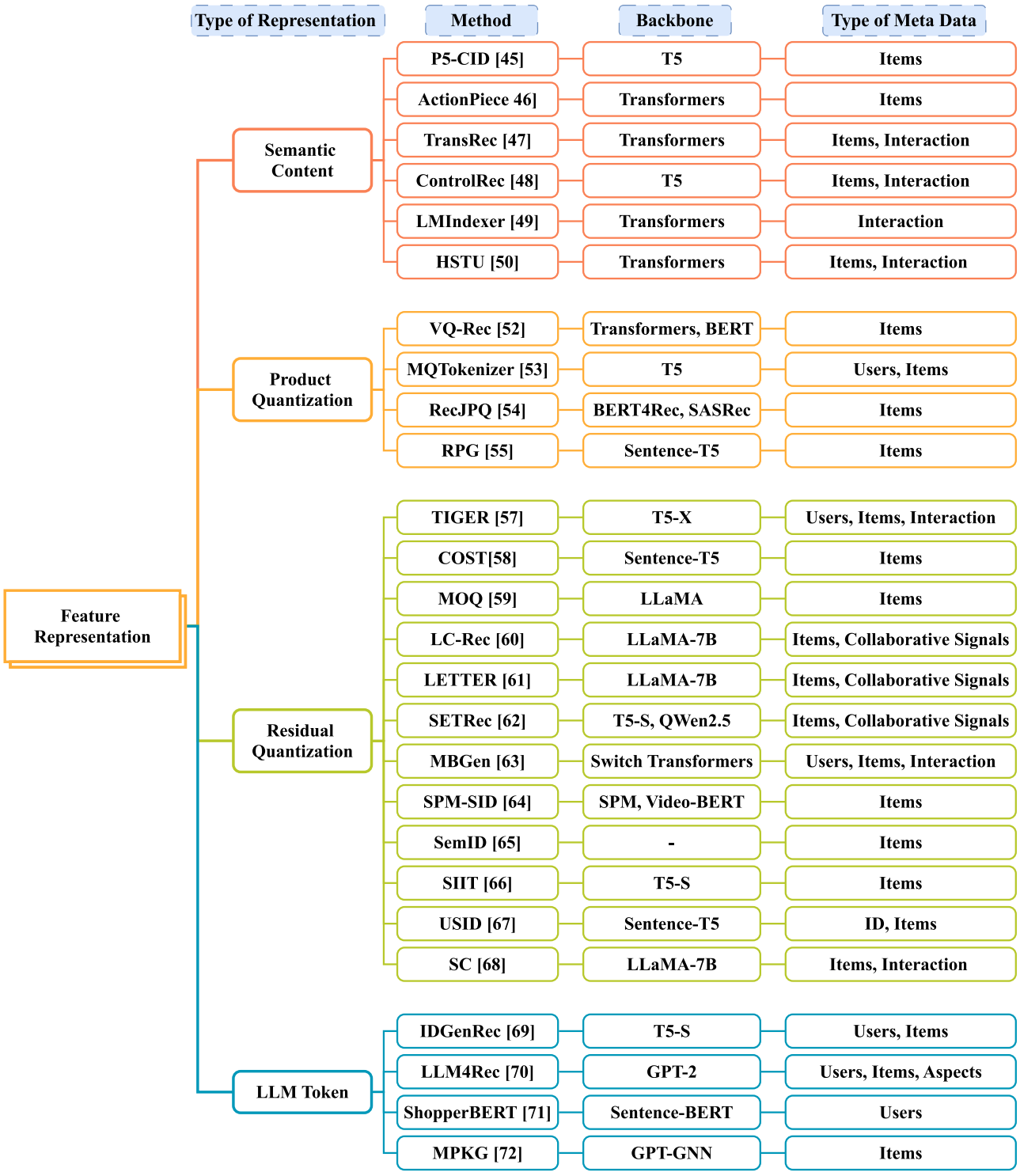


Fig. 2. The classification of feature representation methods. Missing entries are indicated by "-".

2.3.2. Product quantization representation

Product quantization (PQ) [51] has emerged as a prominent compression technique in deep learning for efficient representation of high-dimensional continuous vectors through compact discrete encoding. In recommender system architectures, the PQ framework first embeds items into a continuous vector space, then decomposes these high-dimensional embeddings into multiple mutually independent low-dimensional subspaces. Each subspace undergoes independent quantization processing, concatenating resulting discrete codes constituting a compressed representation of the original item embeddings. This methodology achieves remarkable compression ratios while effectively maintaining the semantic relationships between items in the embedding space, making it particularly effective for large-scale sequential recommendation scenarios.

Since some recommendation tasks cannot directly utilize the continuous semantic embeddings generated by language models, vector quantization effectively transforms these representations into discrete symbol sequences. Hou et al. [52] proposed VQ-Rec, a novel vector

Table 3
Performance comparison of feature representation tokenization methods on Amazon dataset (NDCG@5 and NDCG@10). Missing values are denoted as " - ".

Methods	NDCG@5 on Amazon Dataset						
	Beauty	Sports	Toys	Instruments	Games	Arts	CDs
P5-CID [45]	0.0356	0.0227	-	-	-	-	-
ActionPiece [46]	0.0340	0.0205	-	-	-	-	0.0359
ControlRec [48][1]	0.0431	0.0356	0.0649	-	-	-	-
LMIndexer [49]	0.0262	0.0142	0.0268	-	-	-	-
TransRec [47]	0.0365	0.0360	0.0268	-	-	-	-
RPQ [55]	0.0381	0.0216	0.0401	-	-	-	0.0338
TIGER [57]	0.0321	0.0181	0.0371	-	-	-	-
LC-Rec [60]	-	-	-	0.0856	0.0560	0.0828	-
LETTER [61][2]	0.0355	-	-	0.0789	-	-	-
SETRec [62][3]	0.0370	-	0.0382	-	-	-	-
SIIT [66][4]	0.0367	-	-	0.0797	-	-	-
USID [67]	0.07079	0.02717	0.03141	-	-	-	-
SC [68]	-	-	-	0.0879	0.0591	0.0863	-
IDGenRec [69]	0.0486	0.0326	0.0481	-	-	-	-
	NDCG@10 on Amazon Dataset						
	Beauty	Arts	Instruments	clotheing	Scientific	Pantry	Books
HSTU [50][5]	-	-	-	-	-	-	0.0257
VQRec [52]	-	0.0703	0.0758	-	0.0643	0.0293	-
MQTokenizer [53]	0.0222	-	-	0.0108	-	-	-

[1] ControlRec-B(S), [2] LETTER-LC-Rec, [3] SETRec-QWen1.5B-Cold,
[4] LETTER + SIIT, [5] HSTU-large

quantization-based item representation method for transferable sequential recommendation tasks. Their approach leverages a pre-trained language model to encode descriptive text of items and constructs a mapping from textual encodings to discrete codes using optimized product quantization. By looking up code embeddings based on these indices, new item representations can be efficiently generated, effectively decoupling items from their textual descriptions. To address the challenge of handling massive numbers of users and items in real-world recommender systems, Qu et al. [53] introduced the mask quantization tokenizer (MQTokenizer), which quantizes user-item mask representations learned from collaborative filtering into discrete tokens, enabling the representation of a large number of users and items using a relatively small vocabulary size. As a result, MQTokenizer significantly enhances the tokenization generalization capability of generative recommender systems. Aleksandr et al. [54] proposed a Joint Product Quantization framework called RecJPQ, which learns to represent item embeddings using a limited set of shared subitem embeddings. Instead of relying on full item embedding tensors, RecJPQ concatenates these compact sub-embeddings to approximate the original representations. This approach significantly reduces model size and computational overhead, making it particularly effective for large-scale sequential recommendation scenarios. However, relying solely on limited-length ID sequences may constrain the model's semantic expressiveness. To address this, Hou et al. [55] proposed a parallel semantic ID generation framework for recommendation called RPG. This approach employs an optimized product quantization strategy to encode each item into a long, unordered semantic ID. By incorporating multi-target prediction (MTP) into the training objective, RPG explicitly integrates sub-item semantics into the learning process, enabling the parallel generation of semantic IDs that better capture item characteristics and enhance model expressiveness.

2.3.3. Residual quantization representation

Current research commonly employs the Residual Quantization Variational AutoEncoder (RQ-VAE) [56] to obtain semantic tokens for items. The TIGER framework proposed by Rajput et al. [57] was the first to introduce RQ-VAE for semantic tokenization. This framework utilizes a hierarchical residual encoding structure to decompose raw feature vectors into multiple interpretable semantic subspaces. The model learns a codebook at each quantization level that minimizes reconstruction error through variational inference and enhances encoding robustness via probabilistic distribution constraints. As shown in Eq. (1), RQ-VAE first encodes the input $\boldsymbol{x}$ into a latent embedding representation $\boldsymbol{z} = E(\boldsymbol{x})$ using an encoder $E(\cdot)$. At the initial layer ($d = 0$), the residual is defined as $\boldsymbol{r}_0 = \boldsymbol{z}$. For each layer d, a codebook $C_d = \{\boldsymbol{e}_k\}_{k=1}^{K}$ is defined, where K denotes the codebook size. At layer $d = 0$, the residual $\boldsymbol{r}_0$ is quantized by mapping it to the closest embedding in the codebook, and C_0 denotes the index of the first codeword. Similar to the previous layer, the second codeword is obtained by finding the closest embedding in the codebook at layer $d = 1$, based on the residual $\boldsymbol{r}_0$. This hierarchical residual quantization process continues across subsequent layers, progressively refining the representation through successive residual approximations. The architecture of RQ-VAE is illustrated in Fig. 3.

$$\begin{cases} C_0 = \arg\min_i \|\boldsymbol{r}_0 - \boldsymbol{e}_k\|, \boldsymbol{e}_k \in C_0 \\ \boldsymbol{r}_1 = \boldsymbol{r}_0 - \boldsymbol{e}_{C_0} \end{cases} \tag{1}$$

Since item semantic tokens are crucial to the performance of generative recommender systems, Zhu et al. [58] proposed CoST, a contrastive quantization-based semantic tokenization method. Unlike RQ-VAE, CoST incorporates contrastive learning to refine inter-item neighborhood relationships while capturing semantic information. This approach preserves neighborhood information between input embeddings and their reconstructions while enhancing discrimination from other items. Tan et al. [59] introduced a mixture of quantization (MoQ) mechanism, where item embeddings are first decomposed into two blocks after residual quantization and then clustered in parallel. MoQ generates a single unified codebook by integrating multi-view semantic representations, enabling more effective tokenization.

However, when RQ-VAE independently encodes and discretizes item semantic embeddings in generative recommendation settings, it may lead to distributional bias in the semantic token space. This frequently leads to excessive overlap in the latent space representations of high-popularity items, a phenomenon called codebook collapse. As a result, similar items in dense regions tend to receive highly similar discrete representations, weakening the model's ability to distinguish fine-grained differences. To address this issue, Zheng et al. [60] proposed LC-Rec, a self-improving item tokenization method that uses RQ-VAE with unified semantic mapping to generate multi-level representations. LC-Rec assigns meaningful and conflict-free ID mappings to items and directly generates recommendations using a fine-tuned LLaMA-7B without retrieving candidate items. Wang et al. [61] proposed LETTER, a learnable

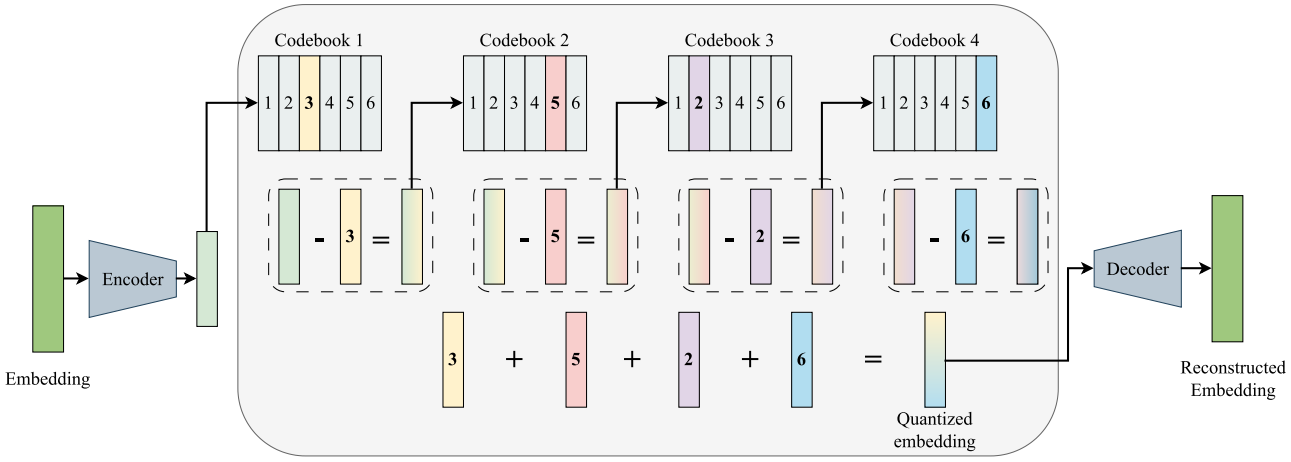


Fig. 3. Overall structure of the RQ-VAE.

tokenizer that integrates hierarchical semantics, collaborative signals, and diversity regularization during code assignment in the RQ-VAE-based token generation process. Additionally, semantic regularization is applied during the semantic vector reconstruction phase. Experimental results based on LC-Rec [60] and TIGER [57] demonstrated LETTER's effectiveness in learning expressive token representations for items. Lin et al. [62] proposed SETRec, which introduces a new identifier paradigm that combines collaborative filtering signals with item semantics to form multi-dimensional item representations. A set of order-invariant tokens represents each item to alleviate local optima issues and improve token generation efficiency in generative recommenders. Liu et al. [63] improved the hierarchical codebook generation process in RQ-VAE to achieve a more balanced token assignment. Singh et al. [64] proposed hashing sub-segments of ID sequences to learn compact discrete representations from frozen content embeddings. Extensive experiments demonstrated that these quantized IDs maintain memorization capability while improving generalization for new and long-tail items. To validate that semantic IDs can enhance the stability of embedding representations, Zheng et al. [65] proposed a semantic ID prefix-ngram approach. This method replaces random assignment with hierarchical clustering based on content embeddings, creating semantic collision hierarchies for items. By doing so, it effectively addresses the issue of embedding instability, improves the modeling of tail IDs, and mitigates representation shift.

To ensure that generated identifiers align with the feature space of LLMs during training, Chen et al. [66] proposed SIIT, a self-improving item tokenization method. SIIT initializes item tokens using RQ-VAE and hierarchical clustering, and allows the LLM to optimize its item identifiers during training periodically. This alignment process ensures consistency between the generated tokens and the LLM's internal understanding of items. Lin et al. [67] proposed a unified semantic and ID representation learning framework that integrates ID tokens and semantic tokens derived from RQ-VAE, balancing unique and shared item characteristics. This approach reduces ID space redundancy while enhancing generalization. Li et al. [68] proposed a token alignment module that transforms item IDs into sequences aligned with the LLM semantic space during the RQ-VAE residual vector encoding stage. Supervised learning aligns collaborative signals with natural language semantics, significantly improving recall performance in generative recommendation.

2.3.4. LLM token representation

With the development of generative recommender systems, constructing token representations that enable the semantic understanding of users and items has become one of the key challenges. Pre-trained language models (PLMs), which have learned rich semantic knowledge from large-scale textual corpora, offer a promising solution by providing a new perspective for addressing this issue. By extracting general semantic features of items through PLMs and converting them into discrete token representations, the modeling capability of recommendation models can be significantly enhanced.

Recent studies have explored integrating LLMs into the token generation process for recommender systems, leveraging their strong contextual understanding and generation capabilities to build more semantically meaningful identifiers for users and items. Tan et al. [69] proposed IDGenRec, where an LLM is used as a semantic identifier generator. It generates semantic IDs based on item metadata, enabling the model to learn and select high-value information identifiers that incorporate both textual and latent semantics. Li et al. [70] introduced LLM4Rec, which utilizes GPT-2 to generate implicit representations of personalized terms from user reviews and concatenates them with standard item embeddings to form expressive feature tokens. Xu et al. [71] proposed ShopperBERT, which directly generates user behavior representations using a language model based on historical interaction data. Fan et al. [72] proposed a pre-training framework for recommendation based on product knowledge graphs, where four pre-training tasks are designed to enable PLMs to generate fine-grained feature tokens for both users and items.

3. Core model frameworks in generative recommender systems

The core objective of generative recommender systems is to generate accurate and personalized recommendations by deeply modeling user behaviors, item content features, and contextual information through generative models. Compared with traditional discriminative recommendation approaches, generative recommender architectures offer distinct advantages in three key aspects: First, they can implicitly model the latent distributions of user-item interactions, thereby effectively mitigating the data sparsity problem commonly encountered in recommendation tasks. Second, by leveraging paradigms such as sequence generation, these models enable end-to-end generation of recommendation lists, enhancing both coherence and flexibility in the recommendation pipeline. Third, integrated with LLMs' semantic understanding and generation capabilities, generative recommenders transcend the reliance on explicit user feedback inherent in traditional collaborative filtering methods. Building upon these strengths, generative recommender systems incorporate heterogeneous data sources, adaptively selecting suitable generative models based on task requirements for training and optimization. This section systematically reviews five major architectural frameworks within this domain: AutoEncoder (AE) models, AutoRegressive (AR) models, Generative Adversarial Networks (GAN) models, Diffusion models, and Large Language Models (LLMs), discussing their key applications and developmental trends in modern

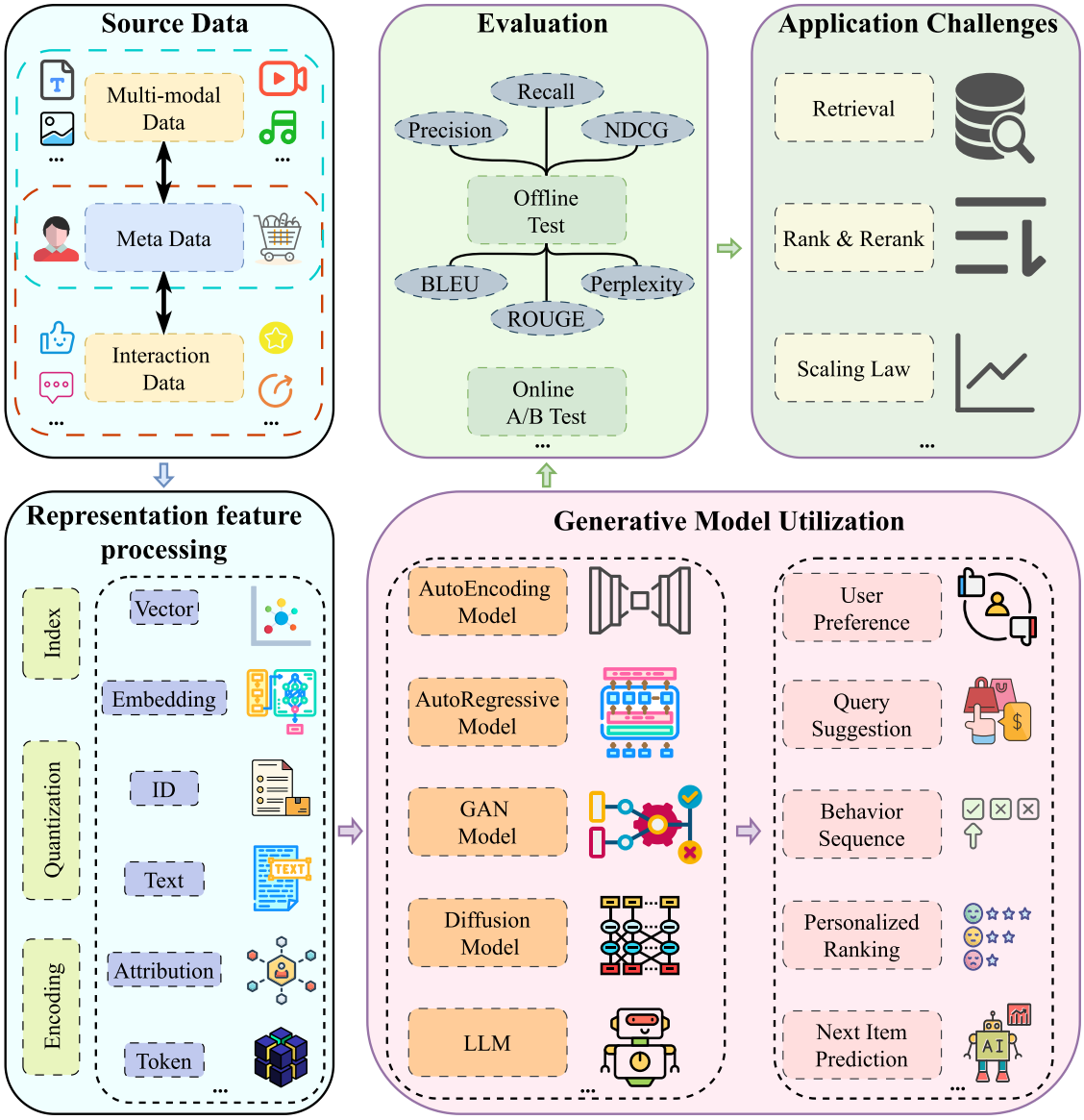


Fig. 4. The general pipeline of generative recommender systems.

recommender systems. The general pipeline of generative recommender systems is illustrated in Fig. 4, which highlights the key processing stages from input to recommendation generation.

3.1. Autoencoding models

AutoEncoder (AE) models are unsupervised neural networks based on an encoder-decoder architecture that learn low-dimensional representations of user-item interactions. The fundamental principle of AE involves reconstructing the original interaction matrix to discover latent user preference patterns. In generative recommender systems, the AE framework processes high-dimensional sparse user interaction vectors through two key transformations: (1) encoding into compact latent representations, and (2) decoding to reconstruct complete preference distributions for personalized recommendation.

Formally, let AE(·) denote the autoencoder function. Given a user-item interaction matrix $\boldsymbol{X} \in \mathbb{R}^{m \times n}$, where m and n represent the number of users and items, respectively. The autoencoder is defined as:

$$\hat{f}(\boldsymbol{x}) = \mathrm{AE}(\boldsymbol{x}, \theta_e, \theta_d) = \mathrm{Decoder}(\mathrm{Encoder}(\boldsymbol{x}, \theta_e), \theta_d), \tag{2}$$

where $\boldsymbol{x}$ denotes the interaction vector of the user, θ_e and θ_d represent the trainable parameters of the encoder and decoder, and $\hat{f}(\cdot)$ is the reconstructed preference distribution used for Top-K recommendation generation.

3.1.1. Denoising autoencoding models

The Denoising AutoEncoder (DAE) [73] is a class of models that learn to recover the original input from its corrupted version. By actively injecting noise into the input data, DAE forces the model to reconstruct the original signal from perturbed or partially damaged observations, thereby enhancing its robustness to noise and capability in feature extraction. Wu et al. [74] proposed the Collaborative Denoising AutoEncoder (CDAE), incorporating DAE into collaborative filtering frameworks and improving generalization performance on Top-K recommendation tasks. In addition, several studies [75,76] have explored the use of Stacked Denoising AutoEncoders (SDAE), where multiple DAE layers are stacked hierarchically to learn multi-level feature representations. These models reconstruct item content information to predict click-through rates, further enhancing recommender systems' expressive power and accuracy.

3.1.2. Variational autoencoding models

The Variational AutoEncoder (VAE) [10] is a generative model capable of learning the underlying distribution of data by introducing probabilistic modeling, enabling the generation of new data samples from the learned distribution. A VAE consists of an encoder and a decoder: the encoder maps input data into a low-dimensional latent space, while the decoder samples from this latent representation to reconstruct the original input. Unlike standard autoencoders that reconstruct inputs via deterministic mappings, VAEs employ variational inference to model probability distributions over neighboring points in the latent space, thereby enabling richer data generation capabilities.

In generative recommender systems, VAEs have demonstrated significant advantages in modeling the implicit distributions of user-item interactions, effectively alleviating data sparsity. Several studies [77,78] have integrated VAEs with traditional collaborative filtering (CF), leveraging their ability to model latent user-item interaction patterns from implicit feedback and thus more accurately uncover user preferences within high-dimensional sparse matrices. Additionally, some works [79,80] directly utilize VAEs for feature extraction and recommendation generation, optimizing the likelihood functions of VAEs according to the characteristics of different user-item interaction data to produce better item recommendations. Furthermore, several works [81,82] combine VAEs with knowledge graph-based representation learning, modeling the probability distributions of nodes and relations in the graph structure. These methods can capture higher-order semantic relationships within complex topologies by incorporating graph constraints to regularize the latent space.

3.2. Autoregressive models

AutoRegressive (AR) models are a class of generative models based on the chain rule of conditional probability decomposition, where the core idea is to predict the next element in a sequence based on historical observations. In recommender systems, autoregressive models effectively capture temporal dependencies in user behavior sequences, enabling the generation of personalized recommendation sequences. By recursively modeling sequential transitions, these models can progressively learn the dynamic evolution of user interests, making them particularly suitable for handling time-ordered behavioral data such as clicks, views, and purchases.

Formally, let AR$(\cdot)$ denote the autoregressive model function, which formulates the recommendation task as a sequence generation problem by modeling the conditional probability distribution of user historical interactions. Given a historical interaction sequence $S = i_1, i_2, \ldots, i_t$, the autoregressive recommendation model can be formally defined as:

$$\hat{y}(S) = \mathrm{AR}(S, \theta) = \prod_{k=1}^{t+1} P(i_k \mid i_1, \ldots, i_{k-1}, \theta). \tag{3}$$

In Eq. (3), S represents the input sequence of length t, θ denotes the model parameters, and $\hat{y}(S)$ is the predicted probability distribution over the next item i_{t+1}. The term $P(i_k \mid i_1, \ldots, i_{k-1}, \theta)$ indicates the conditional probability of item i_k given its preceding interaction history.

Based on differences in context encoding mechanisms, mainstream autoregressive approaches in recommendation can be broadly categorized into two types. The first type is based on Recurrent Neural Networks (RNNs), such as Long Short-Term Memory (LSTM) [8] and Gated Recurrent Units (GRU) [11], which propagate historical information through hidden states and have been widely adopted in early sequential recommendation tasks. The second type leverages self-attention mechanisms, exemplified by the Transformer architecture, which models global dependencies and has demonstrated superior performance and scalability in complex recommendation scenarios due to its ability to handle long-range sequences. These two paradigms, each with distinct strengths, constitute the core technical pathways of autoregressive modeling in generative recommender systems.

3.2.1. Recurrent autoregressive models

GRU4Rec [83] was the first to introduce RNNs into session-based recommendation tasks. It employed stacked GRUs to model the sequential dependencies between items, enabling item-to-item recommendation. Quadrana et al. [84] proposed an improved GRU architecture for sequential recommendation, effectively tracking user preferences' temporal evolution. Yu et al. [85] further utilized RNNs to model dynamic user interest representations and global sequential features, demonstrating that RNNs can effectively capture the order information in user interaction sequences.

To better model user interests across different periods, several studies [86,87] applied modified LSTM networks to next Point of Interest (POI) recommendation tasks. These approaches incorporate spatiotemporal context modeling to enable persistent storage of cross-temporal information, significantly improving recommendation performance in scenarios with periodic interest fluctuations.

Addressing the challenge of modeling long-range dependencies in user behavior sequences for sequential recommendation, recent works have explored selective State Space Models (SSMs), introducing structured state tensor update mechanisms. By dynamically adjusting the temporal attention weights in the state transition matrix, these models can more precisely capture global interest correlations across sessions. Mamba4Rec [88] and RecMamba [89], based on SSM, achieve a balanced trade-off between state compression and fine-grained dependency modeling, offering a more scalable and efficient architecture for handling long behavioral sequences in recommender systems.

3.2.2. Self-attentive autoregressive models

The self-attention mechanism serves as a core component of Transformer [6], enabling dynamic weight allocation across different positions in a sequence and thereby allowing direct modeling of long-range dependencies. This characteristic effectively addresses the limitations of traditional RNN-based models, which suffer from sequential computation and struggle to capture distant relationships. It is particularly suitable for mining non-consecutive interest associations in user behavior sequences.

SASRec [90] was among the first works to introduce unidirectional Transformers into sequential recommendation tasks. Stacking multiple multi-head self-attention layers can extract multi-granular features from long-term and short-term user interactions to more accurately predict user preferences. Subsequent works [91,92] further explored how to incorporate temporal dynamics into Transformer-based sequential recommendation, such as modeling time intervals or periodic patterns in user behaviors to enhance personalization. Moreover, several works [93,94] proposed that self-attention can jointly model diverse user behavior types and contextual signals, thereby improving the relevance and diversity of recommendations.

In recent years, with the rapid development of LLMs based on the Transformer architecture, their application in recommender systems has become increasingly prevalent. LLM-driven generative recommendation architectures demonstrate strong semantic understanding and generation capabilities, which will be discussed in detail in the following sections.

3.3. Generative adversarial network models

Generative Adversarial Networks (GANs) are generative models that approximate the real data distribution through adversarial training between a generator and a discriminator. In this framework, the generator synthesizes realistic samples from random noise. In contrast, the discriminator attempts to distinguish generated samples from real ones, enabling both components to co-evolve toward better performance. Within recommender systems, GANs have been applied to model latent user behaviors, where the generator simulates user interaction preferences and the discriminator assesses the authenticity of generated samples based

on real feedback, thereby enhancing the ability to capture complex interaction patterns and generate high-quality recommendations.

Formally, let GAN(·) denote the GAN model function, the generative adversarial recommendation model is formally defined as:

$$\hat{f}(\boldsymbol{z}, c) = \mathrm{GAN}(\boldsymbol{z}, c, \theta_G, \theta_D), \tag{4}$$

where $\boldsymbol{z} \in \mathbb{R}^d$ denotes the input noise vector sampled from a prior distribution, c represents the conditioning information on the user side, and θ_G and θ_D are the parameters of the generator and discriminator, respectively. The output $\hat{f}(\boldsymbol{z}, c)$ represents the generated user preference distribution.

As a milestone in applying GANs to recommendation tasks, IRGAN [95] was the first to introduce adversarial training into collaborative filtering. It proposed a mechanism in which the generator synthesizes negative samples for adversarial learning with the discriminator, significantly improving recall performance for long-tail items and setting the stage for subsequent research. Following this work, many works have explored applications of GANs in different recommendation domains, including collaborative filtering [96–98], POI recommendation [99,100], and cross-domain recommendation [101–103].

Furthermore, some studies have combined GANs with other generative models to enhance modeling capabilities. For example, several works [104,105] integrated GANs with VAEs to address the limited expressiveness of latent features, thereby improving the robustness and diversity of generated samples and reducing bias and overfitting in recommendations. These hybrid approaches continue to expand the technical frontiers of GANs in recommender systems, advancing the development of more sophisticated and effective generative recommendation architectures.

3.4. Diffusion models

Diffusion models (DMs) [12] are generative models based on Markov chains, which generate user preference distributions through a step-by-step denoising process starting from random noise. In the context of recommender systems, DMs first corrupt user interaction data by progressively adding Gaussian noise until the data becomes pure noise in the forward diffusion process. Subsequently, the model learns a reverse denoising process to reconstruct the original interaction patterns, generating high-quality recommendations.

Formally, let DM(·) denote the diffusion model function. Given a user interaction vector $\boldsymbol{x}$, a diffusion-based recommender can be defined as:

$$\hat{f}(\boldsymbol{x}_0) = \mathrm{DM}(\mathbf{X}_T, c, \theta). \tag{5}$$

In Eq. (5), $\mathbf{x}_T \sim \mathcal{N}(0, I)$ denotes the pure noise input at time step T, c represents the user-side conditioning information, and θ denotes the parameters of the denoising neural network. The output $\hat{f}(\boldsymbol{x}_0)$ represents the reconstructed user preference distribution, obtained through the reverse diffusion process $p_\theta = (\mathbf{X}_{t-1}, \mathbf{X}_t, c)$, which iteratively refines the noisy input into a structured preference representation conditioned on c.

Although some of the generative recommendation methods based on VAE and GAN show some modeling ability of interest, VAE suffers from a posterior collapse problem, and GAN performs unsteadily during adversarial training, which restricts their further development in recommendation tasks. In contrast, as a new generative modeling paradigm, the diffusion model can better model complex distributions and reconstruct interaction features. In recommender systems, the central task is to predict items that users may be interested in based on their historical interactions. This goal aligns closely with the data reconstruction ability of diffusion models. Motivated by this, recent studies have begun exploring how to adapt diffusion models to recommendation scenarios. Wang et al. [106] proposed the first diffusion-based recommendation model DiffRec, which learns the generative process of user interactions through denoising. Specifically, the method first injects Gaussian noise into the user's interaction history during the forward process and then iteratively recovers the original state using a parameterized neural network during the reverse denoising process, enabling accurate modeling of user preferences. Further innovating in this space, Mao et al. [107] introduced DiQDiff, incorporating semantic vector quantization to guide the diffusion process for sequential recommendation, where contrastive divergence maximization enables personalized generation that significantly improves returns in heterogeneous and data-scarce environments.

Traditional collaborative filtering (CF) models exhibit limited capability in extracting collaborative signals, often failing to generate high-quality latent representations in complex scenarios. To address this, Walker et al. [108] pioneered integrating the denoising diffusion probabilistic model (DDPM) into CF frameworks, enabling more effective modeling of user-item interactions. Their approach generates enhanced collaborative signals and learns more robust latent representations. Moreover, several works [109,110] have further explored the incorporation of diffusion models into CF paradigms by treating user-item interaction histories as latent diffusion variables and reconstructing actual preference distributions through backward denoising, demonstrating their potential to overcome the limitations of conventional methods. To address the limitations of static item embeddings in capturing diverse user preferences, Li et al. [111] proposed DiffuRec, the first work to employ diffusion models for sequential recommendation. DiffuRec generates sequence-aware item distributions and injects uncertainty, overcoming the rigidity of conventional fixed vector representations. Other works [112,113] leveraged user behavior sequences as conditional information to guide the denoising process, enabling more temporally coherent generative recommendations.

3.5. Large language models

Large Language Models (LLMs), leveraging vast pre-trained knowledge, strong semantic understanding, and powerful generation capabilities, have introduced a new paradigm for generative recommender systems. LLMs can unify user-item information and interaction signals into natural language sequences and generate recommendation lists or interactive responses through autoregressive mechanisms, breaking the dependency of traditional methods on explicit behavioral data. With significant improvements in semantic reasoning and contextual awareness in recent years, LLMs have evolved from auxiliary feature encoders to core components of end-to-end generative recommendation models.

Formally, let LLM(·) denote the LLM function. Given a user's interaction history X, item features Y, and task prompt T, the LLM-based recommender system is formally defined as:

$$\hat{f}(x) = \mathrm{LLM}(X, Y, T, \theta), \tag{6}$$

where θ represents the LLM parameters, and $\hat{f}(\cdot)$ is the probability distribution of the recommendations generated. As shown in Fig. 5, these approaches can be broadly categorized into two paradigms: non-parametric tuning and parameter tuning.

3.5.1. Non-parametric tuning

Non-parametric tuning methods directly leverage the pre-trained capabilities of LLMs, using input design as the core driver for recommendation generation without updating model parameters. Due to their minimal computational overhead, these approaches offer significant advantages in low-resource scenarios such as few-shot, zero-shot recommendations, and resource-constrained environments. The key idea lies in stimulating the semantic reasoning and knowledge generalization capabilities embedded within LLMs through carefully designed inputs. However, their performance is inherently constrained by the domain coverage of the pre-training corpus and the quality of prompt engineering. Based on differences in input guidance strategies, non-parametric tuning can be further categorized into two main paradigms: direct inference and prompting.

Direct inference does not rely on additional parameter adjustments or task-specific instructions but instead generates recommendations

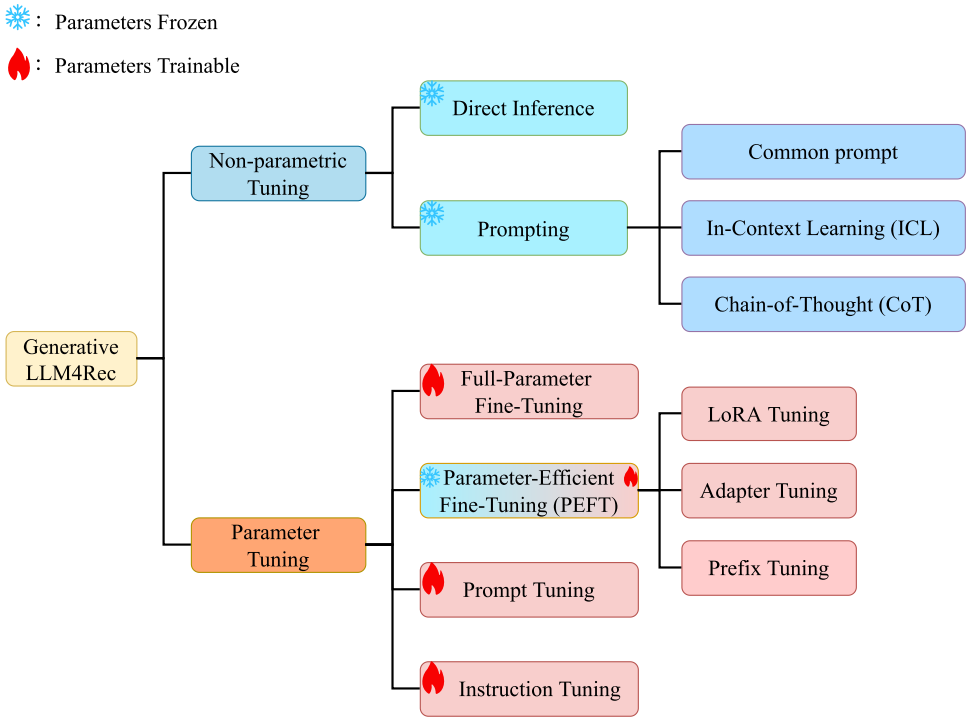


Fig. 5. The classification of the generative LLM4Rec.

directly based on the original pre-trained parameters of the LLM. Typically, these methods follow the pre-training paradigm of LLMs. For example, PTUM [114] introduced two pre-training tasks, including behavior masking prediction and future k-step behavior prediction, to perform user modeling using large-scale unlabeled user interaction data. M6-Rec [115] proposed text filling and autoregressive generation tasks, strictly following the "next token prediction" (NTP) paradigm of LLMs to decode item sequences. P5 [116] employed a multi-task masked language modeling framework, mixing various recommendation datasets for pre-training and transferring the learned representations to downstream tasks. RecGPT [117] represented the first LLM specifically adapted for the recommendation domain, achieving comprehensive domain adaptation through extensive pre-training on a large-scale dataset collected from recommendation-related sources.

Prompting involves explicitly guiding the model's output for specific tasks while keeping the LLM parameters frozen. Based on the form of prompting content, this approach can be further divided into three categories: pure prompting, in-context learning (ICL), and chain-of-thought (CoT) prompting.

Common prompting triggers the LLM's generation capability through direct task descriptions. Several studies [118,119] have designed domain-specific prompt templates tailored to different recommendation tasks to enhance ChatGPT's adaptability in generating relevant outputs. Sanner et al. [120] evaluated the effectiveness of various prompt combinations involving item information and user preferences. Some works [18,121] also utilize LLMs to generate synthetic training data based on existing data and prompt instructions for downstream recommendation models. However, vanilla prompts exhibit limited expressiveness when handling complex user intents.

In-context learning (ICL) injects a small number of task-related examples into the input, enabling the model to implicitly learn the recommendation logic from the input-output mappings of these examples, thereby enhancing situational awareness in downstream tasks. Few-shot or zero-shot LLM-based recommendations using ICL have been applied in diverse domains such as conversational recommendation [122,123], sequential recommendation [124], and personalized recommendation [125,126]. Nevertheless, the performance of ICL heavily depends on the quality and distribution consistency of the provided examples.

Chain-of-thought (CoT) prompting requires the model to explicitly decompose the intermediate reasoning steps in making a recommendation decision, constraining the generation direction through step-by-step outputs. Corresponding reasoning paths must be designed depending on the specific recommendation task and objective. Some studies have applied CoT prompting to sequential recommendation [127,128] and personalized recommendation [129,130]. However, CoT demands stricter logical coherence in prompt design, and increased complexity in the reasoning chain may lead to reduced generation efficiency.

3.5.2. *Parameter tuning*

Parameter tuning adapts LLMs to recommendation tasks by adjusting their weight parameters, aiming to achieve an optimal balance between computational efficiency and model performance while enhancing the modeling of complex recommendation logic. Parameter tuning can be categorized into four major approaches based on differences in parameter scope and optimization strategies.

Full fine-tuning involves end-to-end updates of all model weights to capture domain-specific data distributions. Some studies [131,132] have applied full fine-tuning to pre-trained LLMs using specialized recommendation datasets, enabling effective adaptation to downstream tasks. However, this approach has high training costs, catastrophic forgetting, and potential bias issues.

Parameter-efficient fine-tuning (PEFT) achieves performance comparable to full fine-tuning by updating only a subset of parameters or introducing additional small learnable modules, significantly reducing computational overhead. Recent works have leveraged PEFT to lower the fine-tuning cost of LLMs in recommendation scenarios. Key methods include low-rank adaptation (LoRA) tuning [133–135], adapter tuning [1,136], and prefix tuning [137], which enable more efficient alignment between LLMs and recommendation tasks.

Prompt tuning implicitly adjusts the model generation capability by appending learnable prompt vectors to input sequences. Unlike static prompt-based guidance, prompt tuning introduces new trainable tokens

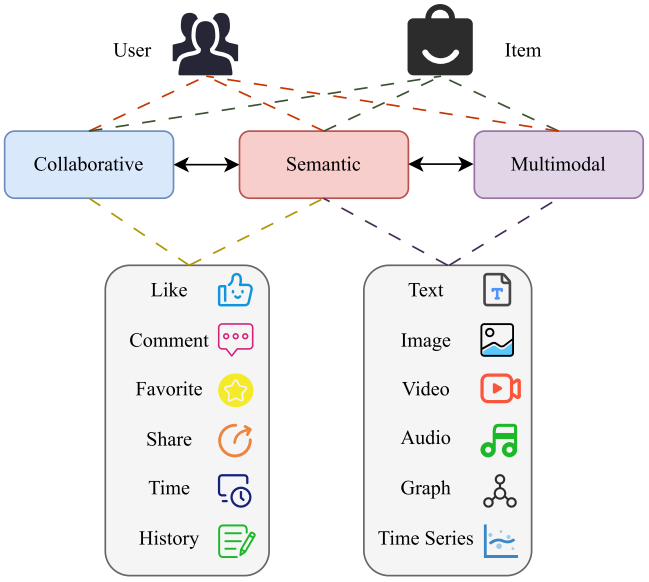


Fig. 6. The main information sources and forms of generative recommender systems.

to optimize model representations dynamically and automatically learns task-specific semantic patterns. Several studies [138,139] have employed hard or soft prompts to guide LLMs in generating desired outputs for downstream recommendation tasks.

Instruction tuning enhances the model's ability to follow complex recommendation logic through explicit instruction-based fine-tuning. Unlike prompt tuning, which focuses on improving performance on specific tasks, instruction tuning typically involves two stages: instruction generation and model fine-tuning, aiming to enhance generalization across unseen tasks. Research [69,140] has shown that instruction tuning significantly boosts LLM performance on targeted recommendation tasks and supports strong zero-shot transfer capabilities.

4. Information fusion methods in generative recommendation

With the evolution of generative recommender systems, information fusion-based approaches aim to integrate multiple information sources to construct more comprehensive user interest representations and item feature characterizations, thereby enhancing recommendation performance. As illustrated in Fig. 6, these sources include collaborative signals from user-item interactions such as likes, comments, favorites, and multimodal features from items such as text, images, videos, and audio, both of which are fused with semantic representations to form the core of generative recommendation. This chapter examines two primary fusion paradigms: (1) multimodal information fusion, and (2) collaborative-semantic information fusion, along with other hybrid techniques. Together, these complementary approaches advance the state of information fusion in generative recommender systems.

4.1. Bridging modality gaps: multimodal information fusion

Multimodal information fusion integrates heterogeneous data modalities, including user behavior data, text, images, videos, and audio, to construct more comprehensive user interest representations and item feature characterizations, enabling more accurate personalized recommendations. In generative recommender systems, this multimodal integration effectively addresses the limitations of unimodal feature representations while providing enhanced semantic understanding and generation capabilities. These systems establish a unified multimodal embedding space through cross-modal representation learning and carefully designed fusion strategies. The core objectives are threefold: (1) bridge semantic relationships between different modalities; (2) create a cohesive representation space; (3) facilitate effective information interaction and integration between modalities. The summary of multimodal information fusion methods is shown in Table 4.

The integration of multimodal information is important to the performance of generative recommender systems, as the choice of fusion mechanism directly governs the depth of cross-modal interaction and the robustness of the final representation. To systematically address this, fusion strategies are broadly categorized into three stages: early fusion, intermediate fusion, and late fusion, each operating at distinct phases of the data processing pipeline and offering unique trade-offs between model complexity, semantic richness, and robustness to data.

- Early fusion integrates raw or low-level features from all modality embeddings at the input stage, creating a unified feature representation before model processing. This approach facilitates rich cross-modal interactions and enables the model to learn intricate inter-modal dependencies from the outset. However, it is highly sensitive to data misalignment and missing modalities, as all modalities must be present and synchronized.
- Intermediate fusion processes each modality through dedicated encoders to extract high-level embeddings before progressively merging them at middle network layers. This paradigm strikes an effective balance, preserving modality-specific semantics while enabling flexible and deep cross-modal interaction. Its main drawback lies in its architectural complexity and the increased computational cost associated with designing and training sophisticated fusion modules.
- Late fusion operates at the decision level, where each modality is processed independently, and prediction scores or embeddings are combined. This method offers high modularity and robustness to missing data. Nevertheless, it often fails to capture fine-grained cross-modal correlations, as the opportunity for feature-level interaction is lost.

Generative recommender systems aim to directly generate candidate item identifiers, while most existing approaches rely on prior knowledge embedded in pre-trained language models to enhance recommendation performance, which typically fail to incorporate multimodal information learning of recommended items. To address this limitation, Luo et al. [141] proposed QARM, a multimodal quantized recommendation framework that first employs pre-trained multimodal large language models (MLLMs) to compress textual, visual, and audio information into unified representations aligned with real user-item interaction distributions, then quantizes these aligned representations into learnable semantic IDs using RQ-VAE for downstream task training, demonstrating significant improvements in online advertising and e-commerce revenue on the Kuaishou platform. Similarly, Zhai et al. [142] introduced MQL4GRec, which transforms and aligns cross-domain items containing both text and image modalities into unified quantized representations sharing a common vocabulary, while incorporating sequential and asymmetric generation tasks to enable knowledge transfer through pre-training and fine-tuning. Complementing these approaches, Wei et al. [143] introduced UniMP, a unified multimodal personalization system that converts item images, text, and IDs into consistent contextual elements using pre-trained vision-language models, thereby enhancing both multimodal understanding and personalized recommendation capabilities. These advances establish effective paradigms for integrating multimodal information into generative recommender systems.

To better integrate multimodal recommendation information, Wei et al. [144] employed graph convolutional networks for modality information aggregation and propagation. However, this approach was not specifically applied in recommender systems. Addressing this limitation, Liu et al. [145] proposed the MMGRec model, which introduces a graph-structured generative paradigm for multimodal recommendation. Their approach utilizes a hierarchical quantized Graph RQ-VAE to assign identifiers to items based on multimodal features, including visual, textual, and auditory information and collaborative interaction signals, while extending semantic tokens with a popularity token to ensure

Table 4
Summary of multimodal information fusion methods.

Fusion Stage	Method	Modal Type	Model
Early Fusion	VIP5 [1]	Text, Image	CLIP, Transformer
	QARM [141]	Text, Video	MLLM
	UniMP [143]	Text, Image	ViT-L, RedPajama
	MMGRec [145]	Text, Video, Audio	Transformer
	InteraRec [146]	Text, Image	GPT4-V
	MLLM-MSR [147]	Text, Image	LLaVA, LLaMA3
	NoteLLM [148]	Text, Tag	LLaMA2
	NoteLLM-2 [149]	Text, Image	MLLM
	DiffMM [151]	Text, Video, Audio	Graph Diffusion
Intermediate Fusion	MVGAE [81]	Text, Image	VAE
Late Fusion	MQL4GRec [142]	Text, Image	LLaMA, ViT-L, T5
	MCDRec [150]	Text, Image	Diffusion

identifier uniqueness, ultimately enabling autoregressive prediction of recommended item identifiers. Further advancing this direction, Yi et al. [81] developed a graph-based multimodal Variational AutoEncoder (MVGAE) that separately models uncertainty differences between semantic features of user/item nodes and various modalities. Their framework fuses modality-specific latent representations into feature-based joint hidden representations, leveraging complementary modality information to enhance model learning capability while achieving robustness against recommendation interaction sparsity and item feature noise.

With the advancement of LLMs, recent studies have explored leveraging prompt engineering to enhance recommender systems by harnessing LLMs' capabilities. Geng et al. [116] proposed the P5 framework, which converts interaction data, item features, and user profiles into a unified natural language sequence, integrating them into personalized prompt templates as model input. This approach employs a single model and objective for multi-task pre-training, enabling seamless adaptation to diverse downstream recommendation tasks. Building upon the P5 paradigm, Geng et al. [1] further incorporated multimodal information, introducing Visual P5 (VIP5), a unified framework for multimodal recommendation tasks. By designing multimodal personalized prompts, VIP5 aligns textual, visual, and user-specific inputs into a shared tokenization space. It employs adapter-based fine-tuning for efficient parameter optimization, achieving superior recommendation performance and training efficiency.

The advancement of MLLMs has significantly enhanced the capability to extract and integrate features from diverse modalities. Karra et al. [146] proposed capturing screenshots during user shopping interactions and leveraging predefined prompts with MLLMs' visual understanding to generate textual preference representations from these screenshots, thereby improving personalized recommendation performance. Building on this, Ye et al. [147] developed an MLLM-enhanced multimodal sequential recommendation model that first converts image features into text using MLLMs, then employs a user summarizer to capture dynamic preference changes, and finally fine-tunes the MLLM-based recommender through supervised learning, demonstrating superior performance in tracking and adapting to evolving user preferences. Further extending this direction, Zhang et al. [148] introduced NoteLLM for item-to-item recommendation, incorporating LLM-generated note summarization and employing instruction tuning to produce thematic tags automatically. Furthermore, NoteLLM-2 [149] implemented prompt-based end-to-end fine-tuning on MLLMs and proposes two novel approaches for balanced multimodal attention: a multimodal in-context learning strategy and an early fusion method that directly integrates visual information into feature representations.

Beyond MLLMs, diffusion models have emerged as a promising solution for multimodal information fusion, leveraging superior generative capabilities and probabilistic modeling advantages. Unlike conventional approaches, diffusion-based recommendation methods enhance generalization performance in data-sparse scenarios. Jiang et al. [150] proposed MCDRec, a multimodal conditional diffusion model that utilizes explicit uncertainty injection to model and fuse multimodal information, integrating pre-extracted multimodal knowledge into item representations while employing diffusion-guided graph denoising to filter user interactions, thereby narrowing the feature space gap between multimodal content features and collaborative signals. Advancing this direction, Ma et al. [151] developed DiffMM, which reconstructs user-item interaction graphs through multimodal graph diffusion while employing cross-modal contrastive enhancement to capture users' modality-specific preferences, ultimately improving recommendation performance through modality-consistent representation learning.

While various generative recommendation methods employ different multimodal fusion strategies as discussed above, they all confront a fundamental challenge: multimodal feature alignment. The effectiveness of these fusion mechanisms critically depends on establishing semantic correspondences between heterogeneous feature representations within a unified latent space. This alignment challenge is particularly pronounced due to the inherent differences in dimensionality, distributional properties, and semantic granularities across modalities such as textual and visual. When modal alignment is inadequate, even sophisticated fusion strategies may introduce semantic inconsistencies, ultimately degrading recommendation quality. Conversely, well-aligned features enable fusion mechanisms to exploit complementary cross-modal information effectively, thereby creating an alignment-fusion synergy that is essential for generating coherent, contextually appropriate recommendations capable of reflecting the complex interplay between diverse user preferences and item characteristics.

4.2. Collaborative and semantic information fusion

In generative recommender systems, the collaborative signals derived from user-item interactions and the semantic information from item content exhibit strong complementary characteristics. While models relying solely on collaborative signals often struggle with data sparsity and cold-start problems, those based exclusively on semantic information may fail to leverage interaction patterns fully. Collaborative signals effectively capture collective preference patterns across user groups, whereas semantic information provides crucial content features and contextual understanding of items. Current approaches predominantly utilize LLMs to obtain textual or visual representations of items, effectively harnessing the models' world knowledge but overlooking the ID-based representations that encode collaborative signals. This limitation particularly degrades recommendation quality when items' multimodal features are sparse. The summary of collaborative and semantic information fusion methods is shown in Table 5.

Recent studies have incorporated pretrained models to extract collaborative signals to address this feature incompleteness. Wang et al. [152] proposed ColaRec for sequential recommendation, which directly generates item identifiers by integrating LightGCN [2]-encoded collaborative signals pretrained on user-item interaction graphs with hierarchically extracted item representations, enabling joint modeling of both

Table 5
Summary of collaborative and semantic information fusion methods.

Method	Collaborative model	Semantic model	Backbone
ColaRec [152]	LightGCN	T5-S	T5-S
PRECISE [153]	MoE	MoE	Transformer
COLLM [135]	MF, LightGCN, SASRec, DIN	LLM	Vicuna-7B
LETTER [61]	SASRec	LLaMA-7B	LLaMA-7B
SETRec [62]	SASRec	Sentence-T5	T5-S, QWen2.5
EAGER [154]	DIN	Sentence-T5	Transformer
SC [68]	DCCF	LLaMA-7B	LLaMA-7B
CTRL [155]	AutoInt	RoBERTa, ChatGLM	-
PRORec [156]	DIN	LLaMA2-7B	Transformer
RLMRec [158]	LightGCN	text-embedding-ada002	LLM

signal types. Song et al. [153] developed PRECIS using a mixture-of-experts (MoE) architecture that combines LLM-derived semantic embeddings with collaborative behavior embeddings, dynamically adapting to users' target interests through comprehensive behavior modeling. Zhang et al. [135] introduced COLLM, a sequential recommendation framework that aligns pretrained collaborative model encodings augmented with user/item ID prompt fields with LLMs' token space without model modification, offering flexible collaborative information integration. Complementary approaches [61,62] directly inject collaborative embeddings from pretrained SASRec [90] as auxiliary side information into the recommendation models.

Since the semantic information of items and collaborative embeddings originates from distinct feature distribution spaces, simply concatenating them via pretrained models may prevent LLMs from effectively understanding collaborative signals. To better integrate behavioral and semantic signals into a unified generative framework, Wang et al. [154] proposed EAGER, a dual-stream generative recommendation architecture. EAGER employs a shared encoder to extract interaction behaviors and semantic features while utilizing two independent decoders to autoregressively decode behavioral tokens and semantic tokens, thereby enhancing the guiding interaction of semantic information on behavioral representations. Li et al. [68] introduced a method where, after item indexing, predefined prompts are used to jointly fine-tune the LLM with a negative sampling task and three positive prediction tasks, aligning and integrating behavioral and semantic signals into the LLM. Further advancing this direction, Li et al. [155] proposed CTRL, an efficient framework that performs fine-grained alignment between collaborative and semantic signals and integrates them into a lightweight collaborative model. After supervised fine-tuning, CTRL enables efficient online deployment in industrial recommender systems. Xiao et al. [156] developed PRORec, a progressive fusion model for collaborative and semantic knowledge. PRORec first employs a non-generative model to integrate semantic knowledge into collaborative embeddings via cross-modal alignment tasks, then uses intra-modal knowledge distillation to help the generative recommendation model effectively capture and combine knowledge from both semantic and collaborative modalities. To further enhance LLMs' understanding of collaborative signals, Liu et al. [157] transformed user-item interactions into hybrid prompt sequences, which are then mapped to both the LLM's token space and collaborative embedding space for cross-modal fusion. Ren et al. [158] proposed RLMRec, which aligns the LLM's semantic space with collaborative relational signals through cross-view alignment, leveraging mutual information maximization to improve feature representation quality.

4.3. Other information fusion methods

Beyond the two primary information fusion approaches discussed earlier, several studies have explored alternative methods to enhance representation learning in generative recommender systems. Zhang et al. [159] developed a social recommendation framework that performs interactive fusion using multi-graph heterogeneous information, incorporating discrete cross-correlation to fuse multi-hop heterogeneous interaction features within a social relation GAN. Zheng et al. [160] proposed a GAN-based approach for fusing explicit and implicit user information, effectively capturing user preferences through neighborhood information aggregation. Advancing the fusion paradigm, Tang et al. [161] introduced an adversarial fusion framework based on modified variational autoencoders that comprehensively leverages multi-view information to generate enhanced user representations. These approaches jointly contribute to the development of information fusion representations for generative recommendation.

5. Evaluation

To adequately evaluate generative recommender systems, it is essential to move beyond the constraints of traditional single-dimensional metrics and establish an integrated evaluation framework encompassing multiple dimensions, including recommendation performance, generated content quality, and computational efficiency.

5.1. Traditional metrics

Applying generative models to traditional recommendation tasks allows for inheriting and adapting well-established evaluation metrics from conventional recommender systems, ensuring comparability and continuity in performance assessment. Evaluation metrics for recommender systems can be primarily classified based on their evaluation context into two main groups: offline metrics and online metrics. Offline metrics are computed using historical data and typically include: (i) metrics assessing the precision of category matching; (ii) metrics evaluating the correctness of predicted recommendation score. Online metrics, in contrast, are measured through live user interactions in production environments and focus on reflecting the capability of traffic conversion in commercial recommendation settings.

5.1.1. Offline metrics

(i) Accuracy of category matching: The primary goal of a recommender system is to ensure the alignment between recommended items and user preferences. Metrics for measuring the accuracy of category matching in recommendations include Precision, Recall, Normalized Discounted Cumulative Gain (NDCG), Mean Reciprocal Rank (MRR), F1 Score, ROC curve, and AUC value.

Precision is the ratio of correctly classified positive samples to the total number of samples classified as positive, calculated as:

$$\text{Precision} = \frac{\sum_{u \epsilon U} |R(u) \cap T(u)|}{\sum_{u \epsilon U} |R(u)|}, \tag{7}$$

where $R(u)$ and $T(u)$ denote the recommended and ground-truth item sets for user u, respectively. U is the set of all users in the evaluation set. Precision is valuable for measuring the proportion of relevant items among recommended ones, making it suitable for scenarios aiming to minimize irrelevant recommendations. However, it overlooks the total pool of relevant items not included in the recommendation list, thus failing to reflect overall coverage.

Recall measures the proportion of actual positive samples that are correctly identified, reflecting the model's ability to cover relevant items, calculated as:

$$\text{Recall} = \frac{\sum_{u\epsilon U} |R(u) \cap T(u)|}{\sum_{u\epsilon U} |T(u)|}, \tag{8}$$

Recall evaluates the system's ability to capture all relevant items, which is particularly useful when broad user interest coverage is desired. However, it may include many irrelevant items to maximize recall, potentially degrading user experience.

F1-Score is the harmonic mean of Precision and Recall, which is defined as:

$$\text{F1}-\text{Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{9}$$

F1-Score balances the two and is appropriate for applications where both high-quality and comprehensive recommendations are important. However, it is less sensitive to extreme values and may not fully represent specific aspects of performance.

NDCG is a metric used to measure the quality of ranking, suitable for ranking problems in recommender systems, calculated as:

$$\text{NDCG} = \frac{\text{DCG}}{\text{IDCG}}. \tag{10}$$

In Eq. (10), $\text{DCG} = \sum_{i=1}^{k} \frac{real_i}{log_2(i+1)}$ and IDCG is the ideal DCG. NDCG accounts for the ranking quality of the recommendation list, making it highly effective for evaluating the performance of Top-N recommendations. Its downside lies in computational complexity and dependence on the definition of item relevance.

MRR is a metric that measures the ranking capability of a recommender system, calculated as the average of the reciprocal ranks of the first correct result for all queries, calculated as:

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{rank_i} \tag{11}$$

In Eq. (11), $rank_i$ is the position of the first correct recommendation. However, MRR only considers the position of the first relevant item and disregards any subsequent relevant items, thus failing to account for the overall coverage or completeness of recommendations.

The **ROC curve** is a graphical tool used to display the performance of classification models at different thresholds. At the same time, the **AUC value** is the area under the ROC curve, indicating the model's overall performance.

(ii) Accuracy of predicted scores: Metrics for measuring the accuracy of predicted scores in recommendations include Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

MAE is the average of the absolute differences between predicted values and actual values, which is calculated as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|. \tag{12}$$

MAE provides a straightforward measure of the average absolute difference between predicted and actual ratings, making it intuitive and easy to interpret. However, it is sensitive to outliers and does not consider the order of recommendations.

MSE is the average of the squared differences between predicted values and actual values, calculated as:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \hat{y}_i\right)^2. \tag{13}$$

MSE amplifies the impact of large prediction errors, which is beneficial for identifying significant deviations; however, it is even more sensitive to outliers than MAE and lacks direct interpretability due to its squared unit.

RMSE is the square root of the MSE, which is calculated as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}. \tag{14}$$

RMSE retains the sensitivity while aligning with the original rating scale, improving interpretability.

5.1.2. Online metrics

Commercial conversion effectiveness: Metrics such as Click-Through Rate (CTR), Conversion Rate (CVR), and Return on Investment (ROI) are inherently online evaluation indicators, designed to measure the real-world business impact of recommendations in live production environments. Unlike offline metrics, these metrics reflect actual user behavior and are typically validated through controlled A/B tests.

CTR represents the probability of users clicking on recommended content, calculated as:

$$\text{CTR} = \frac{\text{Number of Clicks}}{\text{Number of Impressions}}. \tag{15}$$

CTR directly reflects user engagement through click behavior and is widely used in online evaluation. However, it fails to distinguish post-click behaviors and may therefore be misleading.

CVR is the probability of users performing specific actions, such as purchases or registrations after clicking, calculated as:

$$\text{CVR} = \frac{\text{Number of Conversions}}{\text{Number of Clicks}}. \tag{16}$$

CVR measures the likelihood of conversion after a click, making it essential for business-driven recommendation tasks. However, its reliability can be affected by external factors influencing user actions post-click.

ROI measures the return on investment relative to the cost, calculated as:

$$\text{ROI} = \frac{\text{Revenue} - \text{Cost}}{\text{Cost}}. \tag{17}$$

ROI quantifies the economic return from the recommender system and is vital for assessing commercial value. However, it requires precise definitions of cost and revenue, which are often difficult to obtain and optimize directly.

5.2. Generative metrics

Traditional recommendation metrics focus on matching user-item interactions but fail to assess the output's semantic quality and logical coherence in generative tasks. To address this issue, NLP-based metrics can be introduced to evaluate generative recommendation models comprehensively.

BLEU (Bilingual Evaluation Understudy) measures surface-level similarity between generated and reference texts via n-gram precision, which is defined as:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{n} w_n \ \log p_n\right), \tag{18}$$

where p_n is n-gram precision, BP is a brevity penalty, and w_n are weights. Unlike traditional metrics, BLEU evaluates literal text matching rather than user clicks or ratings, but it is insensitive to synonyms and semantic variations, potentially underestimating actual content quality.

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) assesses coverage of key information in reference texts, emphasizing recall. Based on the longest common subsequence (LCS), ROUGE-L is defined as:

$$\text{ROUGE}-\text{L} = \frac{(1+\beta^2) R_{lcs} P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}}. \tag{19}$$

In Eq. (19), R_{lcs} and P_{lcs} are recall and precision, and β balances their weights. Compared to traditional metrics, ROUGE emphasizes recall and coverage but does not account for fluency or logical coherence in the generated text.

Perplexity quantifies the uncertainty of a language model in predicting text, reflecting fluency and coherence. For a given sequence of words $W = (w_1, w_2, \ldots w_N)$, Perplexity is calculated as:

$$\text{Perplexity}(W) = \exp\left(-\frac{1}{N} \ \sum_{n=1}^{N} log p(w_n | w_{<n})\right), \tag{20}$$

where N is the total number of words in the sequence, and $p(w_n|w_{<n})$ represents the conditional probability of the n-th word given the preceding words. Lower perplexity values indicate better alignment with natural language distributions. Unlike behavior-based metrics, perplexity evaluates the model's intrinsic generalization capability.

5.3. Other metrics

In evaluating generative recommender systems, beyond traditional and generative metrics, a comprehensive framework should incorporate computational efficiency assessment. To validate practical performance, computational efficiency evaluation for LLMs is equally critical, covering computational resource requirements during training, including GPU memory consumption and training time, along with inference performance metrics such as token generation latency and retrieval response times, as these metrics directly determine industrial feasibility. We can only holistically assess the overall capability of generative recommender systems by integrating generation quality, user feedback, and computational efficiency.

6. Core challenges and key technical methods

As a critical bridge connecting users with information, recommender systems are pivotal in enhancing user experience and driving platform value. With the continuous expansion of application scenarios and increasing technical complexity, integrating generative recommender systems into industry and academia has become increasingly profound. However, this process presents several core challenges, such as cold start problems, computational efficiency bottlenecks, and complex user preferences modeling. To address these issues, researchers have proposed innovative technical approaches from multiple perspectives, covering key stages of the recommendation pipeline from candidate retrieval to ranking optimization and subsequent re-ranking strategies. Moreover, achieving efficient synergy between large-scale data and complex architectures has become challenging as model scales expand. This section will discuss three key aspects: retrieval, ranking & reranking, and scaling law, systematically reviewing their core methodologies and recent advancements within the context of generative recommender systems.

6.1. Retrieval

In recommender systems, the retrieval component faces critical challenges in efficiently screening billion-scale candidates while capturing dynamic user preferences under strict latency constraints. Recent generative approaches have revolutionized this stage through two key innovations: (1) latent space compression techniques like VAE-based probabilistic sampling that dramatically reduce computational complexity by projecting items into low-dimensional spaces, and (2) end-to-end generation paradigms that directly output candidate IDs. Beyond static retrieval, these methods enable adaptive candidate generation through online learning mechanisms that continuously optimize based on real-time user interactions and system feedback, establishing a closed-loop retrieval-user optimization system that balances relevance and efficiency dynamically.

Bin et al. [162] proposed a vector quantization streaming retriever based on VAE, which dynamically assigns and updates balanced indices for items in real-time, significantly improving retrieval efficiency for large-scale industrial recommender systems. Current research predominantly relies on item ID features for modeling, sacrificing the fine-grained semantic information in traditional embeddings. Zhou et al. [163] investigated the necessity of combining generative retrieval with dense retrieval, demonstrating that exclusive use of generative retrieval introduces inherent biases in identifier generation. To address this limitation, Yang et al. [164] proposed COBRA, a framework that strategically combines sparse IDs capturing coarse-grained semantics with dense embeddings representing fine-grained semantic information, then performs weighted score fusion of both retrieval methods to generate optimized ranking predictions. Yang et al. [165] introduced LIGER, a hybrid model that integrates sequential dense retrieval into generative retrieval frameworks, effectively balancing the strengths of both paradigms to enhance recommendation performance for cold-start items.

Building on these hybrid approaches, recent work has further demonstrated the benefits of jointly optimizing retrieval and recommendation tasks. Penha et al. [166] empirically validated that integrating retrieval with recommendation in a generative multi-task framework regularizes item popularity estimation and latent representations, consistently outperforming single-task training. Complementing this, Zhai et al. [50] proposed unifying retrieval and ranking in generative recommender systems through dynamic modeling of user interaction behavior, achieving more coherent end-to-end recommendation generation. Deng et al. [167] proposed OneRec, an end-to-end generative model that unifies retrieval and ranking stages through joint learning optimization in generative recommender systems, replacing traditional cascaded learning frameworks. The framework further integrates an iterative preference alignment module with Direct Preference Optimization (DPO) to enhance the quality of generated recommendations. To improve personalized query recommendations for e-commerce platforms, Guo et al. [168] introduced OneSug, an end-to-end framework for query suggestion generation. Unlike OneRec, which focuses on item-to-item generation, OneSug emphasizes prefix-to-query generation with relevance-aware constraints. The framework first enhances prefix representations using RQ-VAE to generate structured semantic IDs, which are then fed into a unified encoder-decoder architecture to produce contextually relevant queries. To further align with user preferences, OneSug incorporates a hybrid reward mechanism that optimizes ranking performance based on implicit feedback from query interactions. The representative approaches are listed in Table 6.

6.2. Ranking and re-ranking

Industrial-scale recommender systems typically employ a four-stage cascading architecture comprising retrieval, coarse ranking, fine ranking, and re-ranking. In the retrieval stage, the system initially filters tens of thousands of candidates from a hundred-million-scale item pool. The coarse ranking stage then narrows this down to hundreds of candidates, which subsequently undergo precise scoring in the fine ranking stage. Finally, the re-ranking stage optimizes the ordering of dozens of candidates output by fine ranking to generate the final recommendations. However, this multi-stage processing paradigm faces challenges: processing massive candidate sets while adhering to stringent latency requirements. Traditional collaborative filtering approaches and deep learning-based recommendation models often struggle to simultaneously maintain recommendation coverage and computational efficiency.

In the ranking phase, generative recommender systems reformulate the multi-feature ranking task as a sequence generation problem, enabling end-to-end ranking optimization. These systems generate ranking scores or recommendation lists by jointly modeling user profiles, item attributes, and interaction contexts. This approach eliminates the need for manual feature interaction design in traditional methods and achieves adaptive fusion of heterogeneous features. In recent years, remarkable advancements have been witnessed in generative ranking methods based on Transformers and LLMs. Chen et al. [169] proposed the Rankformer architecture, which innovatively leverages global user-item information to generate enhanced features while achieving significant efficiency improvements through linear-complexity computation. To address the domain gap between general-purpose tasks and recommendation tasks in LLMs, Luo et al. [170] advanced this direction with their Recranker model, introducing adaptive sampling for instruction dataset construction and position shifting techniques to mitigate

Table 6
The representative works exploring the generative retrieval of recommendations.

Methods	Company	Task	Architecture
VQ Retriever [162]	ByteDance	Real-time Indexing	VAE
ROGER [163]	-	Unifying Generative and Dense Retrieval	T5
COBRA [164]	Baidu	Unifying Generative and Dense Retrieval	T5
LIGER [165][1]	Meta	Unifying Generative and Dense Retrieval	T5
GEN_{R+S} [166]	Spotify	Unifying Search and Recommendation	T5
HSTU [50][2]	Meta	Unifying Retrieve and Rank	-
OneRec [167]	Kuaishou	Unifying Retrieve and Rank	MoE
OneSug [168][3]	Kuaishou	End-to-End Generative Query Suggestion	BGE

Code Availability: [1] https://github.com/facebookresearch/liger,
[2] https://github.com/meta-recsys/generative-recommenders,
[3] https://github.com/Edgis/OneSug.

LLM's positional bias, enabling unified optimization for multi-task ranking. Kim et al. [171] proposed the SC-Rec framework that learns diverse preference knowledge through heterogeneous item indexing and multi-prompt templates, performing dynamic re-ranking via confidence-consistency fusion.

Unlike previous approaches that treat items as the fundamental unit of sequence generation, Zhai et al. [50] proposed HSTU, which introduces an interleaved sample organization method to integrate user interaction history and item attribute information into a unified behavioral sequence. This sequence is then used to train a generative ranking model in an autoregressive manner. However, this tokenization strategy doubles the input sequence length, increasing computational overhead and reducing model efficiency. To address this issue, Huang et al. [172] proposed GenRank, a generative ranking method for large-scale industrial scenarios that iteratively predicts actions associated with each item while using the items themselves as contextual positional signals to guide the generation process. This paradigm reduces the input sequence length of Transformer attention mechanisms by half, enabling efficient training and inference in large-scale ranking tasks. Furthermore, Guo et al. [173] introduced the Dual-Flow Generative Recommendation Ranking Network (DFGR), which performs interaction modeling using only raw user behavior sequences and a minimal set of basic behavioral descriptions. DFGR significantly reduces computational costs by employing end-to-end token processing while improving training efficiency and inference performance. Meanwhile, Meng et al. [174] introduced Sequential Ordered Regression Transformer-Generator (SORT-Gen), an end-to-end generative re-ranking approach that integrates multi-objective candidate queues, efficient item selection, and diversity mechanisms into model inference, providing a fast online list generation method. Combining Transformer architectures with ordered regression, SORT-Gen accurately estimates multi-objective values for variable-length sublists, significantly improving click-through rates in online business scenarios.

Previous studies [175] have primarily adopted the two-step generator-evaluator paradigm, where the generator produces feasible sequences and the evaluator selects the optimal sequence based on utility estimation to achieve accurate ranking. Building upon this paradigm, Feng et al. [176] proposed a context-aware generative re-ranking network (GRN) to achieve more precise item re-ranking. Yang et al. [177] proposed a Multi Generator-Evaluator (MG-E) framework, which employs multiple generators to collaboratively produce valuable and diverse candidate lists, which are then fed into an evaluator. By integrating multiple models, this approach avoids the risk of individual generators getting trapped in local optima, thereby providing the evaluator with higher-quality options. Moreover, it reduces the number of candidate lists that need to be evaluated, further enhancing the performance of multi-generator re-ranking. Liu et al. [178] introduced a streaming matching-based generative recommendation strategy, GFN4Rec, which employs an autoregressive item selection model to dynamically append items to the list in each iteration based on the current state space. Some studies have explored diffusion models to generate high-quality results in re-ranking tasks. For instance, Lin et al. [179] proposed a discrete conditional diffusion re-ranking framework (DCDR), which extends traditional diffusion models by introducing a discrete forward process with tractable posteriors, while incorporating a conditional reverse process to generate item sequences tailored to expected user responses, thereby enabling efficient inference.

Table 7
The representative works exploring the generative ranking & re-ranking of recommendation.

Methods	Company	Task	Architecture
Rankformer [169][1]	-	Ranking	Transformer
Recranker [170][2]	-	Ranking	LLM
HSTU [50]	Meta	Ranking	-
GenRank [172]	Xiaohongshu	Ranking	Transformer
DFGR [173]	Meituan	Ranking	Transformer
SORT-Gen [174]	Taobao	Re-ranking	Transformer
PIER [175]	Meituan	Re-ranking	MLP
GRN [176]	Alibaba Group	Re-ranking	Ensemble Model
MG-E [177]	Kuaishou	Re-ranking	Ensemble Model
SC-Rec [171]	-	Re-ranking	Language Model
GFN4Rec [178][3]	Kuaishou	Re-ranking	Autoregressive Model
DCDR [179]	Kuaishou	Re-ranking	Diffusion Model
NAR4Rec [180]	Kuaishou	Re-ranking	Non-autoregressive Model
NLGR [181]	Meituan	Re-ranking	-

Code Availability: [1] https://github.com/StupidThree/Rankformer,
[2] https://github.com/sichunluo/RecRanker,
[3] https://github.com/CharlieMat/GFN4Rec.

Autoregressive models face deployment challenges in real-time industrial recommender systems due to their sequential generation nature, which results in a linear increase in inference time complexity with sequence length. To address this issue, Ren et al. [180] proposed a non-autoregressive generative framework called NAR4Rec for re-ranking, which generates all items in the target sequence simultaneously. Their approach employs a sequence-level uncertainty training method to guide generation and improves decoding through intra-list correlation, significantly accelerating inference while meeting the demands of real-time recommender systems. Similarly, Wang et al. [181] introduced a generative re-ranking method called NLGR that uses neighbor lists, which utilizes a sampling-based non-autoregressive generation strategy to flexibly produce optimal lists in combinatorial space, thereby enhancing both the performance and efficiency of re-ranking. The representative approaches are listed in Table 7.

6.3. Scaling law

The scaling properties of LLMs, particularly the established power-law relationship between model size and performance, have raised critical questions about their applicability to recommender systems. These scaling principles present opportunities and challenges in industrial

Table 8
The representative works exploring the scaling law of generative recommendation.

Methods	Company	Task	Max Parameters
HSTU [50][1]	Meta	Sequential Recommendation	1.5 trillion
LUM [182]	Alibaba Group	Industrial Recommendation	300 million
HLLM [183][2]	ByteDance	Sequential Recommendation	7 billion
LONGER [184]	ByteDance	Conversion Rate Prediction	-
SRT [185][3]	Mercado Libre	Sequential Recommendation	-
MTGR [186]	Meituan	Industrial Recommendation	-
GPSD [187][4]	Alibaba Group	Discriminative Recommendation	125 million - 4 billion

Code Availability: [1] https://github.com/meta-recsys/generative-recommenders,
[2] https://github.com/bytedance/HLLM,
[3] https://github.com/mercadolibre/srt,
[4] https://github.com/chqiwang/gpsd-rec.

deployment scenarios, where optimal resource allocation must balance computational costs with model efficacy. To address this, Zhai et al. [50] pioneered a hierarchical sequential transductive unit (HSTU) encoder architecture that successfully demonstrated the validity of LLM scaling laws in recommendation tasks. Their work marked the first industrial-scale implementation of a trillion-parameter generative recommender system, achieving unprecedented performance while maintaining feasible computational requirements. This breakthrough confirms that the neural scaling laws governing LLMs extend to large-scale recommender systems and provides a practical framework for deploying such massive models in production environments where latency and resource constraints are paramount. Extending the investigation of neural scaling laws to recommender systems, Yan et al. [182] proposed a three-stage paradigm for developing Large User Models (LUMs) in industrial recommendation scenarios. Their empirical results demonstrated that scaling the LUM architecture to 7 billion parameters yielded significant performance improvements, particularly in user preference modeling accuracy. This finding substantiates the critical relationship between model capacity and training data volume in recommender systems, as both the LLM parameters and training data scale synchronously, the model's capability to capture nuanced user interests exhibits consistent enhancement. Chen et al. [183] proposed an innovative Hierarchical Large Language Model (HLLM) framework featuring a dual-tower architecture that separately processes item features and user preferences. This novel design, implemented with 700 million parameters, effectively demonstrates the inherent scalability of language model architectures in recommendation scenarios.

Modeling long user behavior sequences while preserving long short-term interest representations remains a critical challenge in recommender systems. To address this, Chai et al. [184] proposed Long-sequence Optimized Transformer for GPU-Efficient Recommenders (LONGER), an end-to-end framework that extends user sequence modeling to 5000 interactions while maintaining computational efficiency. The architecture proposed a token merging mechanism that reduces information loss from raw sequences, and a hybrid attention strategy that cuts FLOPs by approximately 50 percent. Extensive validation on billion-scale industrial offline datasets and multiple online service scenarios demonstrates that LONGER achieves superior modeling capacity and establishes industrial scaling laws for sequence-based recommender systems. Building on the discussion of efficient sequence modeling, Pablo et al. [185] further investigate how to adapt transformer architectures to capture scalable user preference modeling. They replace traditional trainable embeddings with a parameter-efficient feature extractor that decouples model size from catalog dimensions. They also develop a contrastive learning formulation to preserve catalog diversity in high-dimensional preference spaces. The proposed method, SRT, bridges the gap between theoretical scaling potential and practical constraints in high-dimensional sequential recommendation scenarios. To enable the deployment of generative recommender systems in industrial scenarios, Han et al. [186] proposed the Meituan Generative Recommendation (MTGR) framework, which is built upon the HSTU architecture. MTGR employs a multi-sequence encoding mechanism while preserving all information, including features for comprehensive modeling. In addition, the training and inference frameworks of MTGR have explicitly been optimized to meet industrial-scale requirements. The effectiveness of the Scaling Law has been validated through its deployment in the Meituan food delivery recommendation scenario. To mitigate overfitting in large-scale industrial recommendation models, Wang et al. [187] proposed the Generative Pretraining for Scalable Discriminative (GPSD) Recommendation framework. This approach leverages pretraining and a frozen sparse parameter strategy to effectively reduce overfitting in discriminative models. By extending the dense parameter scaling of Transformers, GPSD establishes a scaling law for discriminative recommendation tasks, enabling more efficient and generalizable model development in industrial settings. The representative approaches are listed in Table 8.

7. Challenges and future directions

While generative recommender systems have demonstrated significant advances, their practical implementation and theoretical development continue to face fundamental challenges. These limitations span data, computing, and security dimensions, prompting ongoing research efforts to address their constraints. This section examines these persistent challenges and identifies open research questions demanding further investigation.

7.1. Multi-dimensional information fusion

Generative recommender systems can obtain richer information representations by integrating diverse data types, yet effectively fusing such heterogeneous data remains challenging due to differences in modalities, structures, and semantics [142]. Current fusion methods still require improving semantic consistency and complementary relationships across multimodal data. Furthermore, dynamic user behaviors and real-time data streams further increase fusion complexity. A critical challenge lies in developing fusion approaches that can effectively integrate multi-source data, generate unified representations, and maintain computational efficiency [141]. Future research should focus on two key directions: developing advanced fusion techniques capable of dynamically filtering and combining information according to recommendation requirements while preserving completeness and fidelity, and ensuring semantic consistency during fusion while optimizing real-time performance and scalability to enable dynamic recommendation tasks.

7.2. Hallucination mitigation

While generative recommender systems leverage generative models' world knowledge and natural language capabilities to enhance recommendation quality, autoregressive generation's inherent probabilistic sampling nature often leads to factual and logical hallucinations [188].

These errors can significantly reduce recommendation reliability, particularly in high-stakes domains such as medical diagnosis. Additionally, distributional biases in training data may induce semantic-drift hallucinations [189]. Future research should explore multi-dimensional mitigation strategies through two complementary approaches: integrating dynamic knowledge graphs and external knowledge bases with retrieval-augmented generation (RAG) [19] to provide reliable prior knowledge for constrained generation, while simultaneously employing uncertainty quantification methods to identify and filter low-reliability outputs and mitigate probabilistic generation biases.

7.3. Privacy protection and recommendation safety

Current generative recommender systems face dual challenges in privacy protection and recommendation security [190]. On the one hand, model training requires extensive user data mining, including personal profiles and interaction histories, yet existing privacy protection techniques fail to adequately balance data utility with anonymization strength [191]. On the other hand, generative models remain vulnerable to prompt injection attacks, data poisoning, and discriminatory content generation. Future research should address these challenges through specific approaches. Privacy protection requires developing federated learning frameworks that handle heterogeneous data distributions and implementing adaptive differential privacy mechanisms that dynamically adjust noise levels based on data sensitivity. Recommendation security demands integrating causal debiasing techniques with adversarial training to mitigate bias propagation and developing multi-stage content moderation engines that perform real-time monitoring across the generation process and output results. Ethical alignment necessitates establishing automated review mechanisms that incorporate interpretable human value alignment models and deploying adaptive filtering strategies capable of detecting emerging harmful content patterns across diverse recommendation domains. These methods will enable recommender systems to achieve optimal performance while maintaining strict privacy and safety standards.

7.4. Construction of a novel evaluation framework

Evaluation for generative recommender systems needs to go beyond the metrics traditionally used in conventional recommender systems, particularly regarding the quality, diversity, and personalization of the generated recommendations [19]. Traditional evaluation indicators often fail to capture user satisfaction with generated content and cannot detect potential biases or misinformation embedded within the outputs. In domains such as news and e-commerce, ensuring generated content's factual accuracy and fairness is of particular importance [18]. Therefore, one of the key challenges lies in developing an evaluation framework that effectively integrates traditional recommendation metrics with capabilities specific to generative models. Future research should focus on building more comprehensive evaluation frameworks that align with the evolving nature of generative recommender systems. This includes incorporating diverse generative evaluation methods to complement traditional metrics and actively measuring the presence of misinformation in the generated content to enhance system trustworthiness. Additionally, it is crucial to assess the impact of recommender systems on users' cognitive biases and the formation of information cocoons. By establishing a cross-domain and cross-scenario comprehensive evaluation system, we can provide reliable benchmarks and facilitate the responsible development and deployment of generative recommendation technologies.

7.5. Lightweight generative architecture design

Current generative models, especially LLM-based approaches, commonly face the challenge of high computational costs, which limit their deployment and practical application in resource-constrained industrial scenarios. Traditional large-scale pre-trained models often require substantial computational resources for recommendation tasks, significantly increasing the response latency in online systems [192]. Future research should address specific efficiency challenges through targeted solutions. Model compression requires developing domain-aware pruning algorithms that identify and preserve critical parameters while achieving parameter reduction without significant performance degradation. Knowledge distillation demands designing multi-stage distillation frameworks that transfer both semantic knowledge and specific patterns from teacher models to lightweight student architectures. Dynamic inference optimization requires the implementation of adaptive Mixture-of-Experts (MoE) mechanisms [168] that selectively activate expert networks based on user query complexity and recommendation context. Hardware-aware optimization requires developing quantization strategies tailored to recommendation workloads and designing efficient attention mechanisms that reduce computational complexity. These approaches aim to strike a better balance between model performance and efficiency, facilitating the practical deployment of generative recommender systems in real-world environments.

8. Conclusions

In this paper, we present a comprehensive survey of recent advances and the main pipeline of generative recommender systems, providing in-depth analyses across multiple dimensions, including data source, feature representation, model architecture, information fusion, and evaluation frameworks. The study demonstrates that generative recommendation exhibits remarkable advantages over traditional collaborative filtering methods regarding personalization capability, accuracy, and scenario adaptability. However, three fundamental bottlenecks persist in practical deployment: the limited representation learning capacity caused by inadequate information fusion, the constraints of computational costs and inference latency on real-time services, unresolved challenges in hallucination mitigation, standardized evaluation, and privacy preservation mechanisms. These bottlenecks limit existing models' scalability, reliability, and applicability in real-world scenarios. Based on our comprehensive analysis, we delineate future research directions and provide deeper insights for researchers in this field.

CRediT authorship contribution statement

Yuqiu Zhao: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization; **Chaohong Tan:** Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Data curation; **Lei Shi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization; **Yan Zhong:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization; **Feifei Kou:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization; **Pengfei Zhang:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis; **Wei Chen:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation; **Chen Ma:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the Open Project Program of Guangxi Key Laboratory of Digital Infrastructure (No. GXDIOP2024010), the Fundamental Research Funds for the Central Universities (No. CUC25SG013), the Key Laboratory of Public Opinion Governance and Computational Communication (No. YQKFYB202501), the Key R&D Program of Zhejiang Province (No. 2024C01164) and the Zhejiang Provincial Natural Science Foundation of China (No. LMS25F020005).

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